The Root-Locus Design Method

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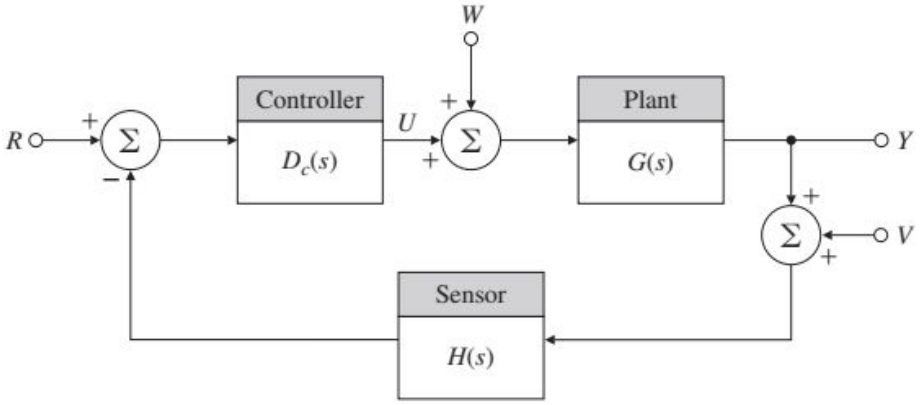
Reference book for the presentation: **Feedback Control of Dynamic Systems, 8th Edition, Franklin • Powell • Emami-Naeini, Pearson, 2019, ISBN: 978-0-13-468571-7**

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1. Root Locus of a Basic Feedback System

1. Put the characteristic equation in a polynomial form
2. Select the parameter of interest **K**
3. We can define component polynomials a(s) and b(s) such that the characteristic equation is a(s) + Kb(s) = 0
4. We then define the transfer function L(s) so that the characteristic equation can be written as =>



$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{D_c(s)G(s)}{1 + D_c(s)G(s)H(s)}$$

$$1 + D_c(s)G(s)H(s) = 0.$$

$$a(s) + Kb(s) = 0. \qquad L(s) = \frac{b(s)}{a(s)}$$

$$1 + KL(s) = 0 \text{ where } L(s) = \frac{b(s)}{a(s)}$$

1. Root Locus of a Basic Feedback System

- Usually **K** is the gain of the controller then:
  - L(s) is proportional to Dc(s)G(s)H(s)
- Evans proposed: to plot the locus of all possible roots of the characteristic equation for $0 < K < \infty$ to determine the best value of K
- Studying the effects of additional poles and zeros on this plot, we can determine the consequences of additional dynamics added to Dc(s) (design of the controller)
- The plot is named **root locus**
- The rules to construct the root locus is called Evan's method

1. Root Locus of a Basic Feedback System

Assuming

- b(s) and a(s) are **monic** (the coefficient of the highest power of s is 1)
- b(s) is degree **m** and a(s) is degree **n** with **m ≤ n**
- The roots of b(s) = 0 are the zeros of L(s) => z_i
- The roots of a(s) = 0 are the poles of L(s) => p_i
- The root of a(s) + Kb(s) = 0 are the poles of the closed-loop system => $\mathbf{r_i}$

$$1 + KL(s) = 0 \text{ where } L(s) = \frac{b(s)}{a(s)}$$

$$\begin{aligned} b(s) &= s^m + b_1 s^{m-1} + \cdots + b_m \\ &= (s - z_1)(s - z_2) \cdots (s - z_m) \\ &= \prod_{i=1}^{m} (s - z_i) \end{aligned}$$

$$\begin{aligned} a(s) &= s^n + a_1 s^{n-1} + \cdots + a_n \\ &= \prod_{i=1}^{n} (s - p_i). \end{aligned}$$

$$a(s) + Kb(s) = (s - r_1)(s - r_2) \cdots (s - r_n)$$

1. Root Locus of a Basic Feedback System

There are different ways to write the characteristic equation of the closed-loop system

- Root-locus form or Evans form
  - Where K > 0
- For each value of K there is a set of roots of the characteristic equation

$$1 + KL(s) = 0 \text{ where } L(s) = \frac{b(s)}{a(s)}$$

$$1 + KL(s) = 0,$$

$$1 + K\frac{b(s)}{a(s)} = 0,$$

$$a(s) + Kb(s) = 0,$$

$$L(s) = -\frac{1}{K}$$

1. Root Locus of a Basic Feedback System

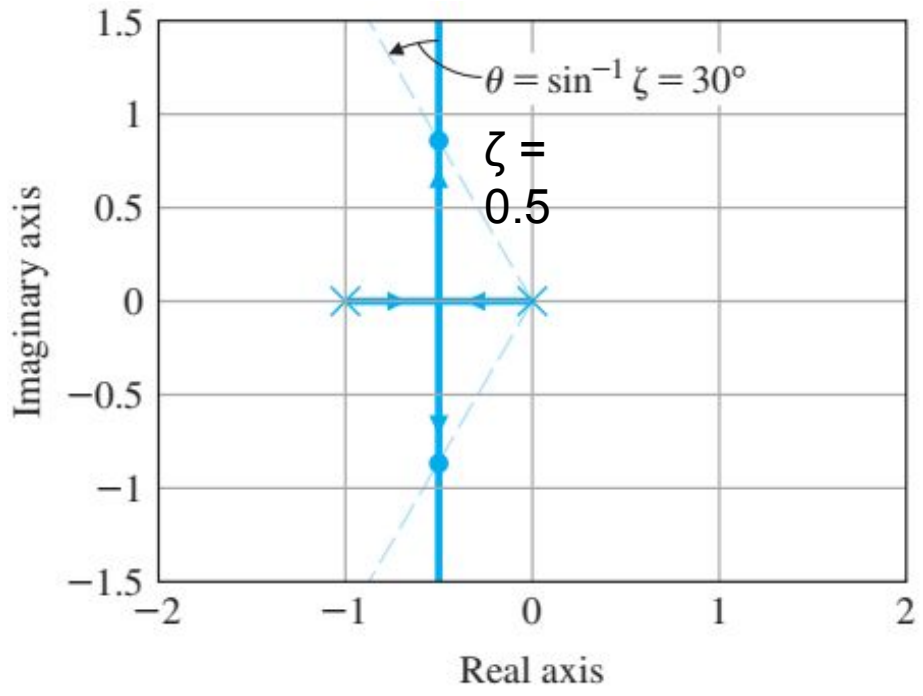
Example: Root Locus of a Motor Position Control

Solve for the root locus of closed-loop poles of the system created by feeding back the output θ_m with respect to the parameter **A** if Dc(s) = H(s) = 1 and also c = 1.

$$\frac{\Theta_m(s)}{V_a(s)} = \frac{Y(s)}{U(s)} = G(s) = \frac{A}{s(s+c)}$$

1. Root Locus of a Basic Feedback System

Example: Root Locus of a Motor Position Control



$$\frac{\Theta_m(s)}{V_a(s)} = \frac{Y(s)}{U(s)} = G(s) = \frac{A}{s(s+c)}$$

$$L(s) = \frac{1}{s(s+1)}, \qquad b(s) = 1, \qquad m = 0, \qquad z_i = \{\text{empty}\},$$

$$K = A, \qquad a(s) = s^2 + s, \qquad n = 2, \qquad p_i = 0, -1.$$

$$a(s) + Kb(s) = s^2 + s + K = 0.$$

$$r_1, r_2 = -\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2}$$

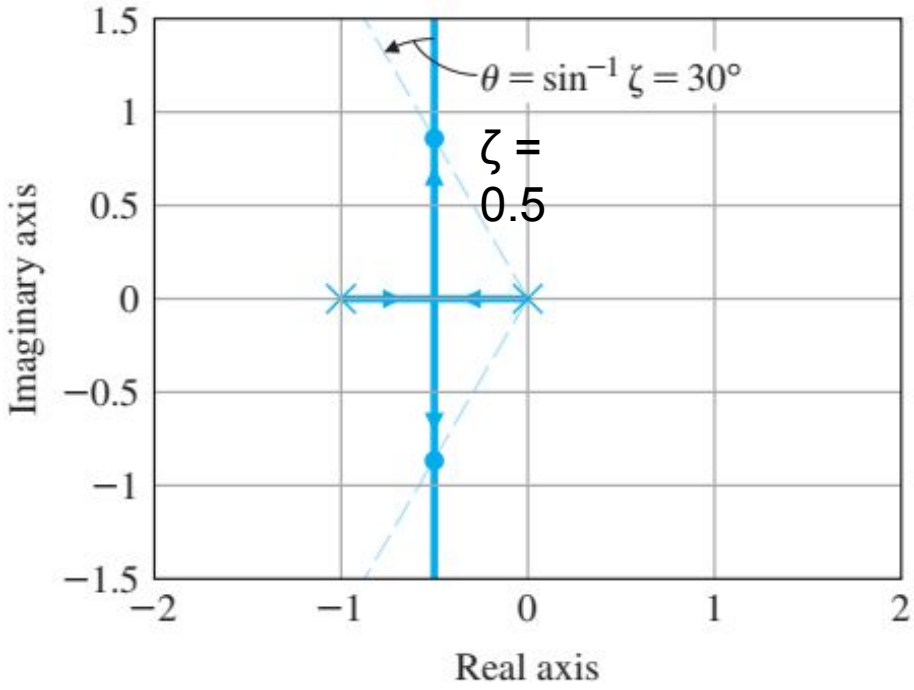
To find w_d for ζ = 0.5 (intersection):
$w_d = (½)/\tan(30°) = \sqrt{3}/2$
Then, **K = 1**

$$\frac{\sqrt{4K-1}}{2} = \frac{\sqrt{3}}{2}$$

1. Root Locus of a Basic Feedback System

Example: Root Locus of a Motor Position Control

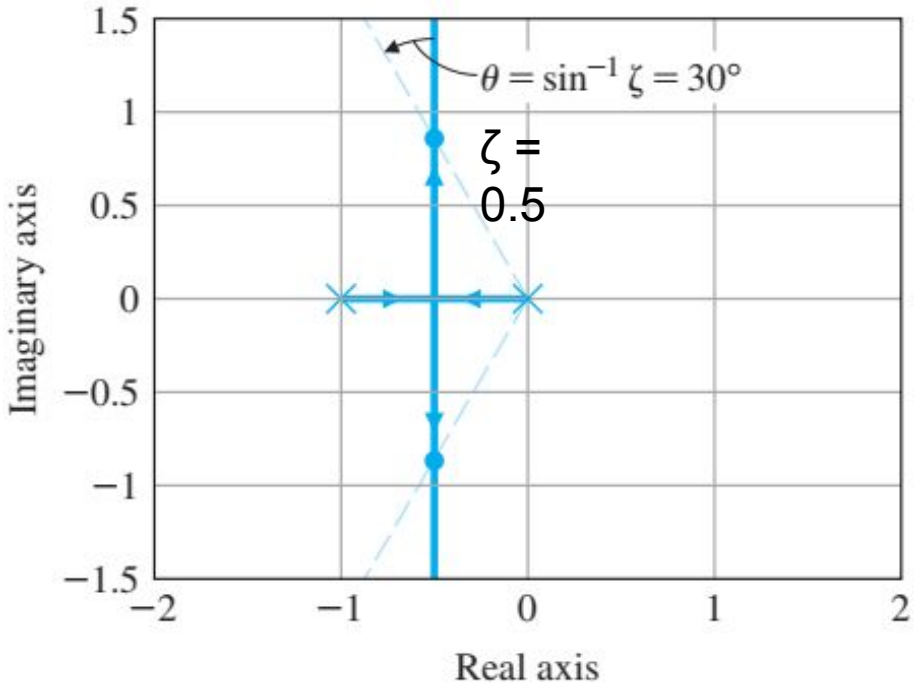
- There are two roots, thus, two loci which we call **branches** of the root locus
- For K = 0 (open-loop) **the branches begin** at 0 and -1 (open-loop poles)
- As K is increased, the roots move toward each other, coming together at s = −1/2 (breakaway point)
- The roots move off to ∞ with equal real parts (the sum is always −1)



1. Root Locus of a Basic Feedback System

Example: Root Locus of a Motor Position Control

- Modifying **K** we change the location of the closed-loop poles but with the same settling time for complex poles
- If the dynamic response is satisfactory => we define K for a desired ζ; otherwise, we require a more complex controller
- The same logic can be applied for other parameters of the system that enters linearly in the characteristic equation



1. Root Locus of a Basic Feedback System

Example: Root Locus with Respect to Plant Open-Loop Pole and Zero

With Dc(s) = H(s) = 1, find the root locus of the characteristic equation with respect to **c**.

$$1 + G(s) = 1 + \frac{s + 2c}{s(s + c)}$$

1. Root Locus of a Basic Feedback System

Example: Root Locus with Respect to Plant Open-Loop Pole and Zero

With Dc(s) = H(s) = 1, find the root locus of the characteristic equation with respect to **c**.

$$1 + G(s) = 1 + \frac{s + 2c}{s(s + c)} \qquad s^2 + s + c(s + 2) = 0.$$

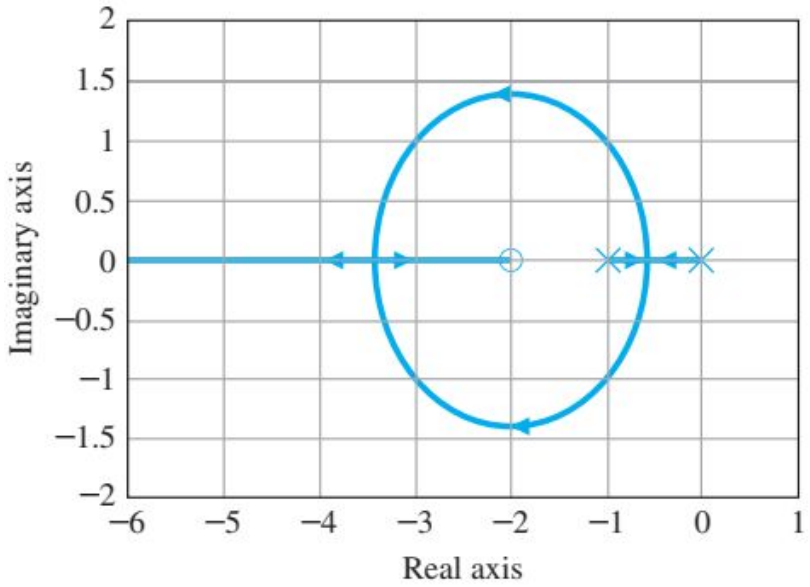
$$L(s) = \tfrac{s+2}{s(s+1)}, \qquad b(s) = s + 2, \qquad m = 1, \qquad z_i = -2,$$

$$K = c, \qquad a(s) = s(s + 1), \qquad n = 2, \qquad p_i = 0, -1.$$

The root-locus form

$$1 + \frac{c(s + 2)}{s(s + 1)} = 0.$$

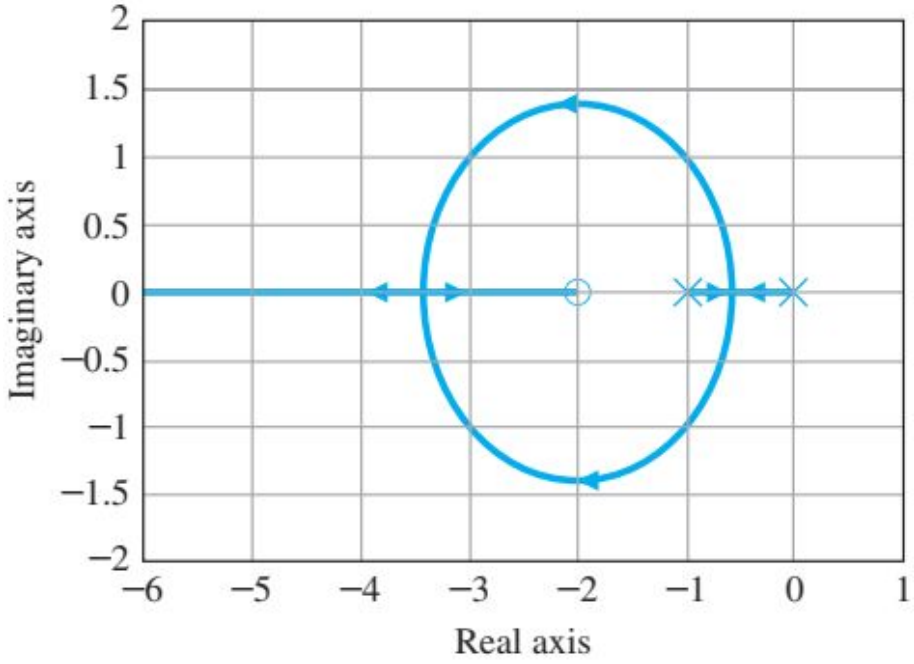
$$r_1, r_2 = -\frac{c + 1}{2} \pm \frac{\sqrt{c^2 - 6c + 1}}{2}$$



1. Root Locus of a Basic Feedback System

Example: Root Locus with Respect to Plant Open-Loop Pole and Zero

- When $c^2 - 6c + 1 < 0$, the roots become complex, when $0.172 < c < 5.828$
- When $c < 0.172$, the roots are on the real axis between $s = 0$ and -1 points
- There are two roots at
  - $s = -0.586$, when $c = 0.172$ (breakaway point)
  - $s = -3.41$ when $c = 5.828$ (break-in point)
- When $c > 5.828$, the two locus segments move in opposite along the real axis
  - One to ∞
  - One to the zero



1. Root Locus of a Basic Feedback System

- Root locus can be done by Matlab (rlocus(G))

In control design we are interested in:

- How a controller modifies the dynamics of the closed-loop system
- Propose a controller that will meet the dynamic response specifications a required closed-loop performance
- Evaluate the correctness of a computer-generated locus
- It is difficult the generate the root locus for more complex systems (when the characteristic equation is order 3 or higher)

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus

Definition I. The root locus is the set of values of s for which $1 + KL(s) = 0$ is satisfied as the real parameter K varies from 0 to $+\infty$. Typically, $1 + KL(s) = 0$ is the characteristic equation of the system, and in this case the roots on the locus are the closed-loop poles of that system.

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus

$$L(s) = -\frac{1}{K}$$

Definition II. The root locus of $L(s)$ is the set of points in the s-plane where the phase of $L(s)$ is 180°. To test whether a point in the s-plane is on the locus, we define the angle to the test point from a zero as ψ_i and the angle to the test point from a pole as ϕ_i then Definition II is expressed as those points in the s-plane where, for an integer ℓ,

$$\sum \psi_i - \sum \phi_i = 180° + 360°(l - 1)$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus

- The importance of Definition II: while it is very difficult to solve a high-order polynomial by hand, computing the phase of a transfer function is relatively easy
- The usual case: **K** is real and positive => positive or 180° locus.
- Other case: **K** is real and negative => negative or 0° locus
- We can determine a positive root locus for a complex transfer function by measuring the phase and marking those places where we find 180◦

$$L(s) = -\frac{1}{K}$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus

Example:

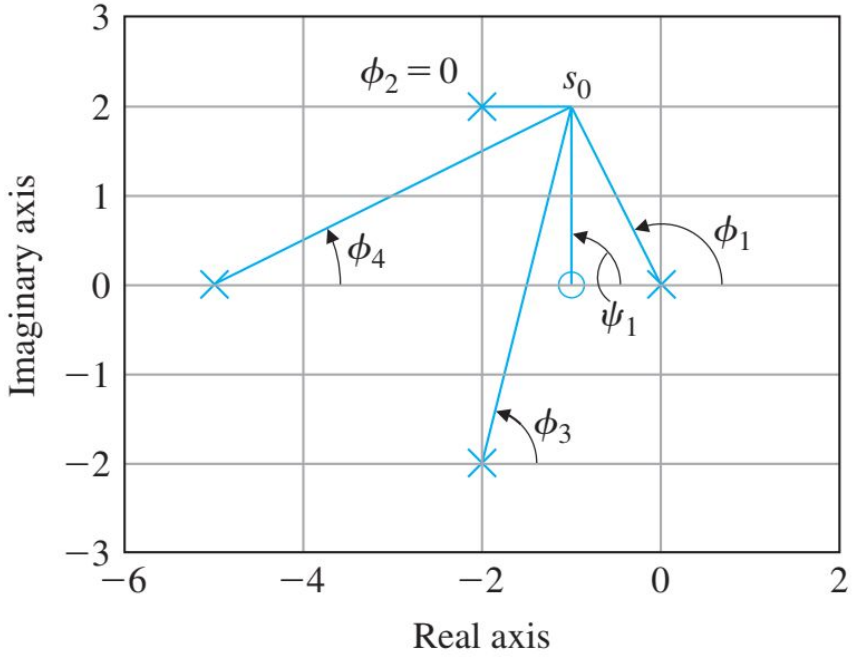
$$L(s) = \frac{s+1}{s(s+5)[(s+2)^2+4]}$$

Suppose we select the test point $s_0 = -1+2j$

Does s_0 lies on the root locus for some value of K?

If so: $\angle L(s_0) = 180° + 360°(l-1)$

$$\angle(s_0+1) - \angle s_0 - \angle(s_0+5) - \angle[(s_0+2)^2+4] = 180° + 360°(l-1)$$



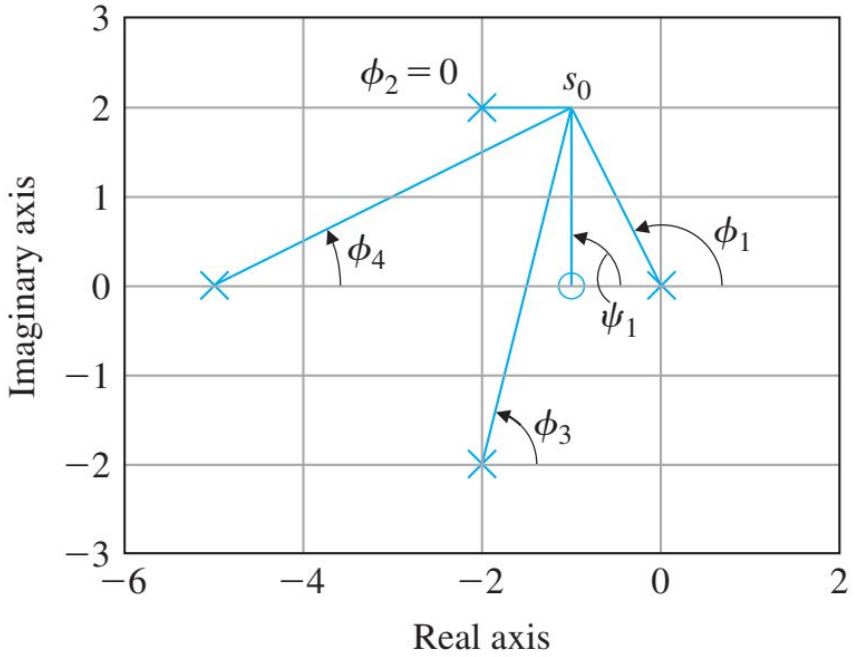
2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus

Example:

- The angle from the zero term $s_0 + 1$ can be computed by drawing a line from the location of the zero at -1 to the test point s_0 => ψ_1 and $\psi_1 = 90°$
- The same approach can be used for the angles from each pole to s_0

$$\begin{aligned}\angle L &= \psi_1 - \phi_1 - \phi_2 - \phi_3 - \phi_4 \\ &= 90° - 116.6° - 0° - 76° - 26.6° \\ &= -129.2°.\end{aligned}$$



- Angle of L(s) is not 180° => **s_0 is not** on the root locus
- Try another point on the plane => **not practical**

We need guidelines

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus

- Evans developed 5 rules
  - Rules 1 – 3 are commonly used
  - Rules 4 and 5 used occasionally

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 1:

The **n branches** of the locus **start** at the **poles of L(s)** and **m** of these **branches end** on the **zeros of L(s)**.

$$1 + KL(s) = 0,$$

$$1 + K\frac{b(s)}{a(s)} = 0,$$

$$a(s) + Kb(s) = 0,$$

$$L(s) = -\frac{1}{K}$$

- If K = 0 => characteristic eq. is a(s) = 0 (poles of L(s))
- When K -> ∞
  - b(s) = 0 or
  - s → ∞
- Since there are **m zeros** where **b(s) = 0**, **m branches** can **end** in those zeros

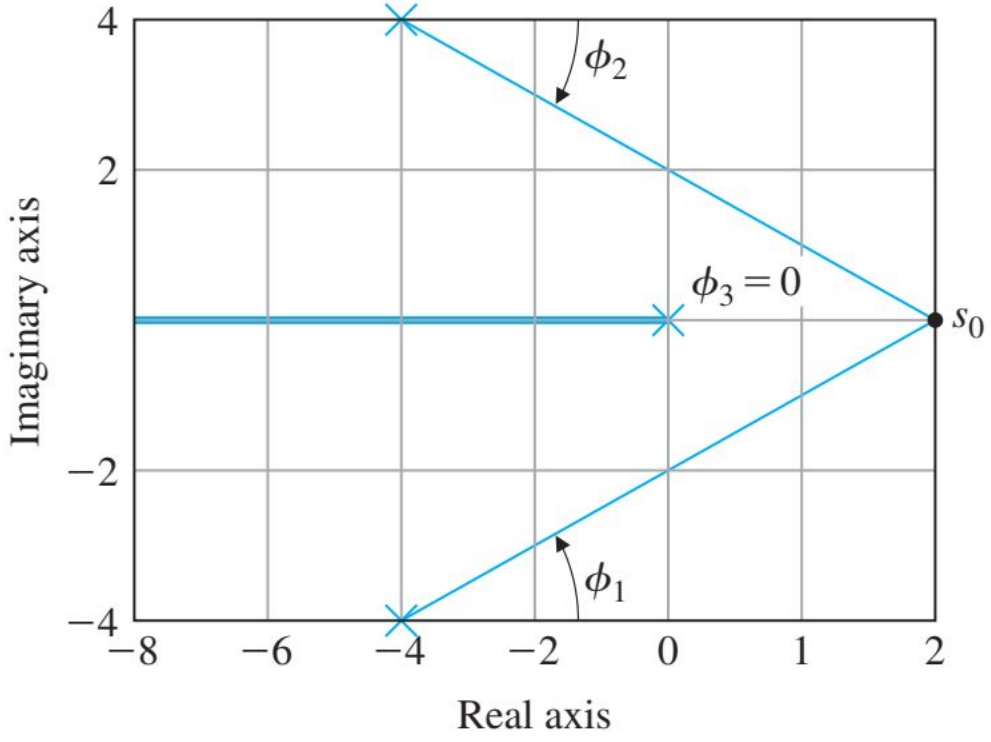
2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 2: The loci are on the real axis to the left of an odd number of poles and zeros.

- For s_0 the angles of the complex poles cancel each other
- Angles from real poles or zeros are
  - 0° if the test point is to the right
  - 180° if the test point is to the left
- An odd number of poles and zeros to the left of a point in the real axis produces:

$$ang\big(L(s)\big) = 180° + 360°(l-1)$$



2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

<u>Rule 3 (for n > m, more poles than zeros in L(s)):</u> For large values of **s** and **K**, n − m branches of the loci are asymptotic to lines at angles φ_l radiating out from the point $s = \alpha$ on the real axis, where

$$\phi_l = \frac{180° + 360°(l-1)}{n-m}, \qquad l = 1, 2, \ldots, n-m,$$

$$\alpha = \frac{\sum p_i - \sum z_i}{n-m}.$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 3 (for n>m): Explanation

- Ch. Eq. L(s) = -1/K for K -> ∞ can be satisfied only if
  - L(s) = 0 (at zeros of L(s))
  - s -> ∞
- L(s) = b(s)/a(s) can be **forced** to approximate the form $L(s) \approx 1/(s-\alpha)^{n-m}$
  - i.e. n-m poles at s = α
  - **Like seen the poles and zeros with a zoom-out**
  - **m** zeros **cancel the effect** of the **n** poles

$$L(s) = -\frac{1}{K}$$

$$(1) \qquad 1 + K\frac{s^m + b_1 s^{m-1} + \cdots + b_m}{s^n + a_1 s^{n-1} + \cdots + a_n} = 0$$

The locus of (1) is asymptotic to the locus of (2) when K -> ∞ and s -> ∞

$$(2) \qquad 1 + K\frac{1}{(s-\alpha)^{n-m}} = 0.$$

$L(s) \approx 1/(s-\alpha)^{n-m}$ can be obtained by calculating a(s)/b(s) and matching the dominant two terms (highest powers in s) to the expansion of $(s-\alpha)^{n-m}$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 3 (for n>m): Explanation

- For $L(s) \approx 1/(s-\alpha)^{n-m}$
- Evaluating if a point $s_0 = Re^{j\varphi}$ is in the locus (with R large and φ variable)
- All the n-m poles produce the same angle φ_l
  - s_0 is in the locus if $(n-m)*\varphi_l = 180° + 360°(l-1)$
  - With $l = 1, \ldots, n-m$
- The lines of the asymptotic locus **come from** $s_0 = \alpha$ on the **real axis**
- How to calculate α?

$$(n-m)\phi_l = 180° + 360°(l-1)$$

$$\phi_l = \frac{180° + 360°(l-1)}{n-m}, \quad l = 1, 2, \ldots, n-m.$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 3 (for n>m): Explanation

- How to determine α?
  - With some properties of polynomials
- Expanding the right side
  - Coefficient of s^{n-1} is $-p_1 - p_2 - \cdots - p_n$
- From the left side
  - Coefficient of s^{n-1} is a_1
- The same analysis can be done for the numerator b(s)
- The sum of the closed-loop roots (r_i) is the negative of the coefficient of s^{n-1}
  - This coefficient is **independent of K** if **m < n − 1**
  - The center point of the roots does not change with K if m < n − 1

$$s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_n = (s - p_1)(s - p_2) \cdots (s - p_n).$$

$$-b_1 = \sum z_i,$$
$$-a_1 = \sum p_i.$$

The characteristic equation can be written as:

$$s^n + a_1 s^{n-1} + \cdots + a_n + K(s^m + b_1 s^{m-1} + \cdots + b_m)$$
$$= (s - r_1)(s - r_2) \cdots (s - r_n) = 0.$$

if **m < n − 1**

$$a_1 = -\sum r_i.$$

Then: $$-\sum r_i = -\sum p_i$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 3 (for n>m): Explanation

- Center point of a monic (coefficient 1 for s^n higher power) polynomial: is the average of its roots (or the centroid or geometric center of the roots in the complex plane)

$$P(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1} s + a_n$$

The center point can be calculated as follows:

Option 1: $$\bar{s} = \frac{1}{n} \sum_{i=1}^{n} s_i$$

Option 2: $$\bar{s} = \frac{-a_1}{n}$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 3 (for n>m):

For **large values of K** the n closed-loop roots (r_i) go to:

- **m roots** (r_i) **approach** the **m zeros** (z_i)
- **n − m roots** (r_i) **approach** the asymptotic system of $1/(s-\alpha)^{n-m}$ (at ∞)

$$-\sum r_i = -(n-m)\alpha - \sum z_i = -\sum p_i$$

Solving α:

$$\alpha = \frac{\sum p_i - \sum z_i}{n-m}$$

Note that $\sum p$ and $\sum$ only require the real parts.

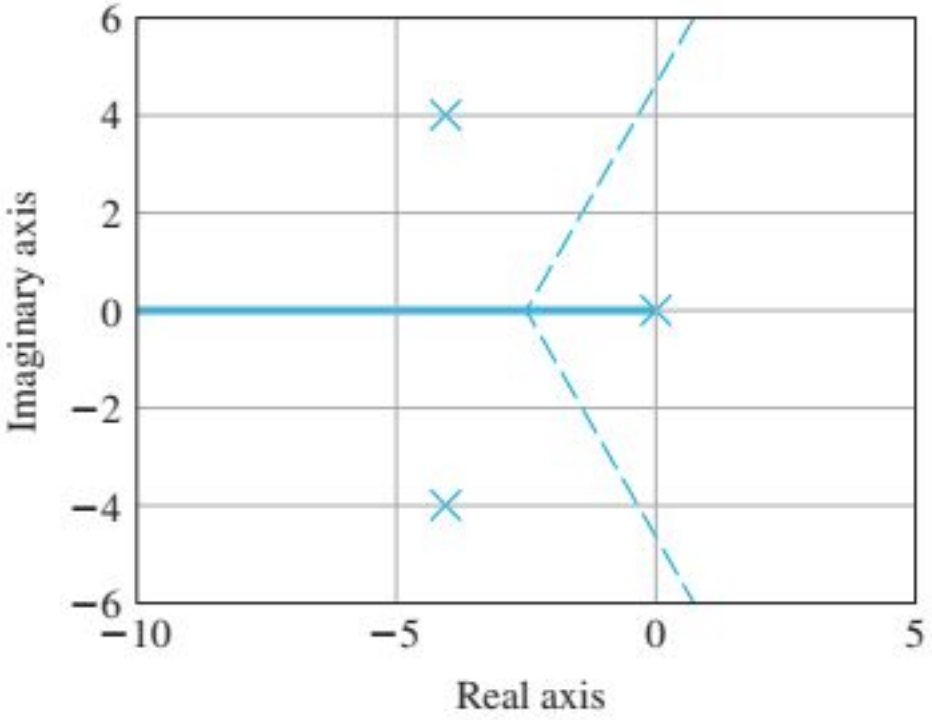
Imaginary parts of the complex conjugate poles cancel eachother.

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus: Example

- **Rule 1:** 3 poles => 3 branches that end in ∞ (0 zeros)
- **Rule 2:** there is locus to the left of an odd number of poles and zeros

$$L(s) = \frac{1}{s[(s+4)^2 + 16]}$$



2. Guidelines for Determining a Root Locus

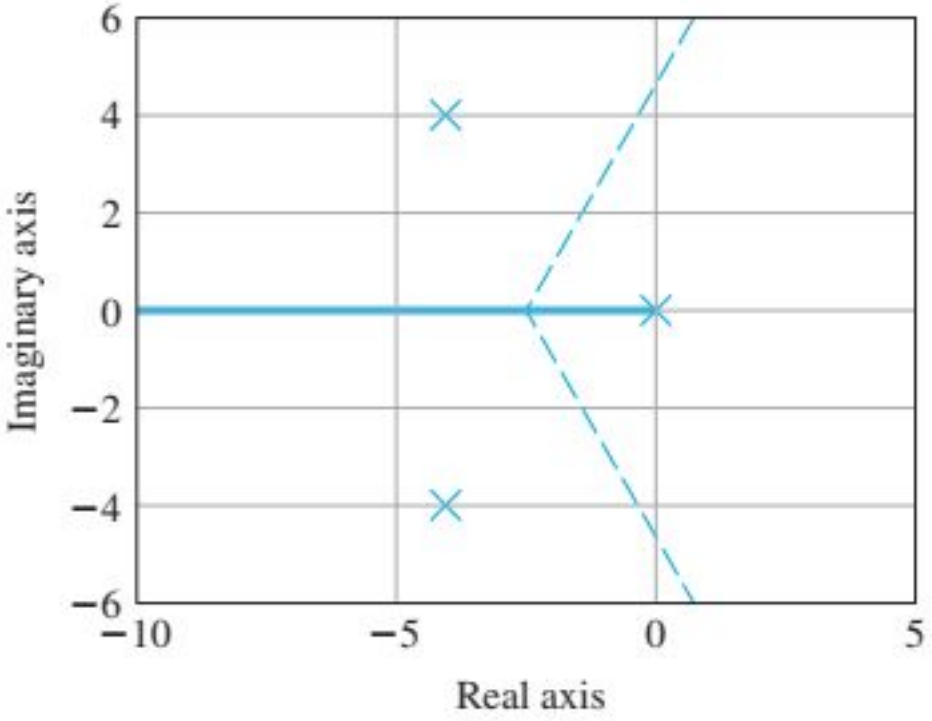
Guidelines for Determining a Root Locus: Example

$$L(s) = \frac{1}{s[(s+4)^2 + 16]}$$

- **Rule 3:** calculate α (starting point of the asymptotes)
  - p_i: 0, -4+4j, -4-4j
  - z_i: no zeros
  - n = 3, m = 0

$$\alpha = \frac{\sum p_i - \sum z_i}{n - m} \qquad \alpha = \frac{(0 - 4 + 4j - 4 - 4j) - (0)}{3 - 0}$$

$$\alpha = \frac{-8}{3} = -2.67$$



2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus: Example

$$L(s) = \frac{1}{s[(s+4)^2 + 16]}$$

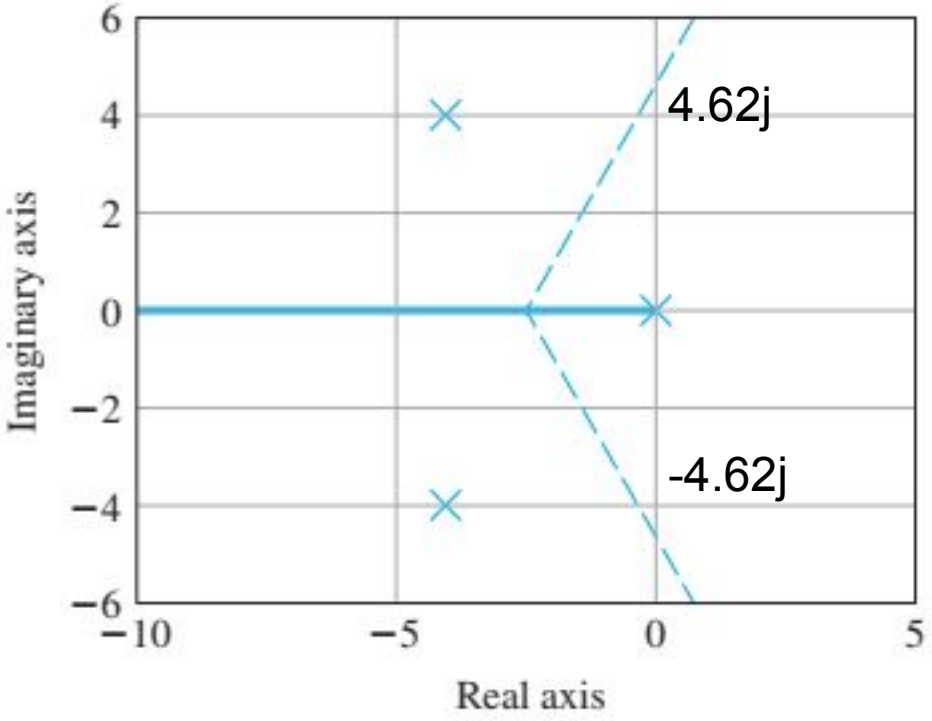
- **Rule 3:** calculate the angles of the asymptotes
  - n = 3, m = 0
  - l = 1, 2, 3

$$\phi_l = \frac{180° + 360°(l-1)}{n-m}, \quad l = 1, 2, \ldots, n-m.$$

$$\phi_{l1} = \frac{180° + 360°(1-1)}{3-0} = 60°$$

$$\phi_{l2} = \frac{180° + 360°(2-1)}{3-0} = 180°$$

$$\phi_{l3} = \frac{180° + 360°(3-1)}{3-0} = 300° = -60°$$

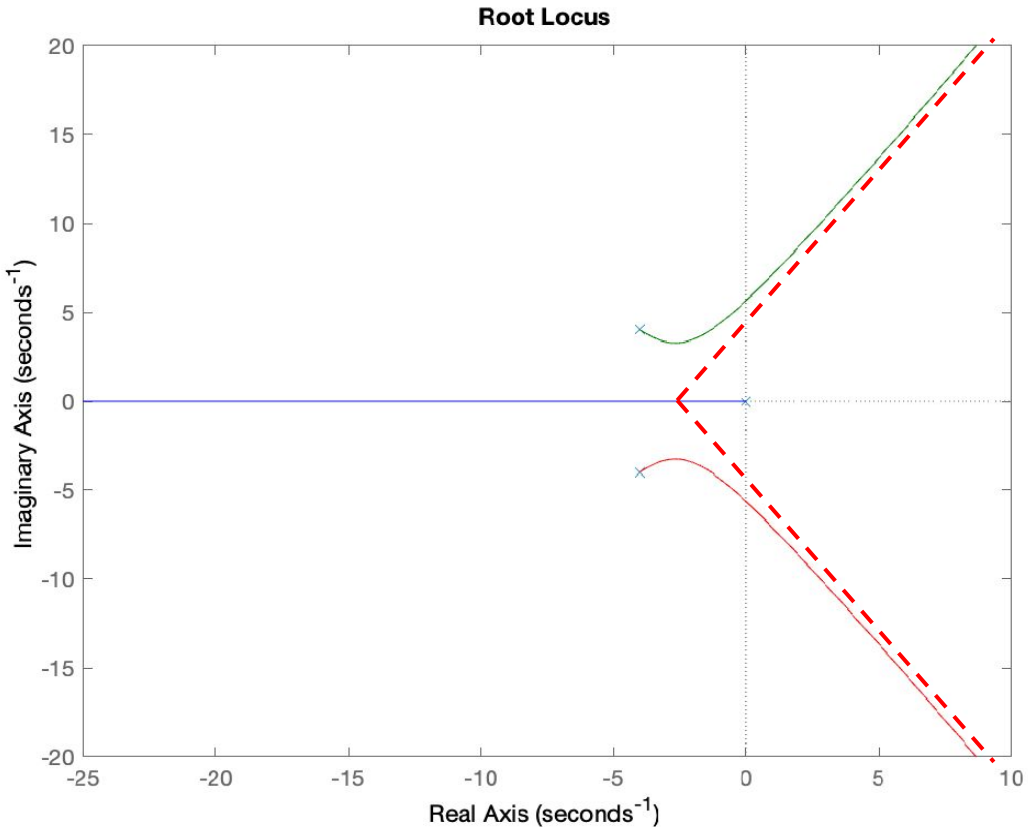


2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus: Example

- **Matlab: rlocus(G)**

```
s=tf('s');
G=1/(s*((s+4)^2 + 16));
rlocus(G);
```



2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 4: The angle of departure of a branch of the locus from a single pole is given by the equation.

$$\phi_{dep} = \sum \psi_i - \sum_{i \neq dep} \phi_i - 180°$$

Excep the analized pole (the one we are calculating the angle of departure)

Sum of the angles **to** the **remaining** poles $\sum \psi_i$

Sum of the angles **to all** the zeros $\sum \phi_i$

The angles of departure for repeated poles with **multiplicity, q**, is given the equation.

$$q\phi_{l,dep} = \sum \psi_i - \sum_{i \neq l,dep} \phi_i - 180° - 360°(l-1)$$

l is an integer l = 1, 2, ... , q.

q branches depart from a pole with **multiplicity q**

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 4: The angle(s) of arrival of a branch at a zero with **multiplicity q** is given by the equation

$$q\psi_{l,arr} = \sum \phi_i - \sum_{i \neq l,arr} \psi_i + 180° + 360°(l-1)$$

Excep the analized zero (the one we are calculating the angle of arrival)

Sum of the angles **to all** the poles

$$\sum \psi_i$$

Sum of the angles **to** the **remaining** zeros

$$\sum \phi_i$$

l is an integer l = 1, 2, ... , q.

q branches arrive to a zero with **multiplicity q**

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

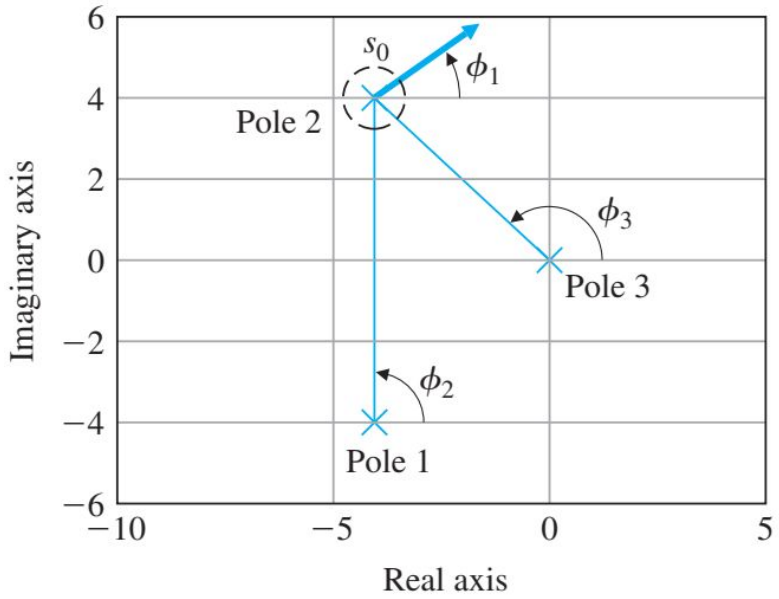
(1) $$\sum \psi_i - \sum \phi_i = 180° + 360°(l - 1)$$

$$L(s) = \frac{1}{s[(s+4)^2 + 16]}$$

Rule 4: Example

- Consider a test point s_0 **very near** the **pole x** (analyzed)
- The angle from pole x to s_0: φ_1
- Considering the example
  - (1) must be fulfilled for a point in the locus
  - l = 1 (multiplicity of the poles)
  - Leaving φ_1 to the right of (1) and the rest of the terms to the left

$$\begin{aligned}\phi_1 &= -90° - 135° - 180° \\ &= -405° \\ &= -45°.\end{aligned}$$



2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 5:

- The locus can have multiple roots at points on the locus and the branches will approach a point of **q** roots at angles separated by

$$\frac{180^\circ + 360^\circ(l-1)}{q}$$

- and will depart at angles with the same separation

2. Guidelines for Determining a Root Locus

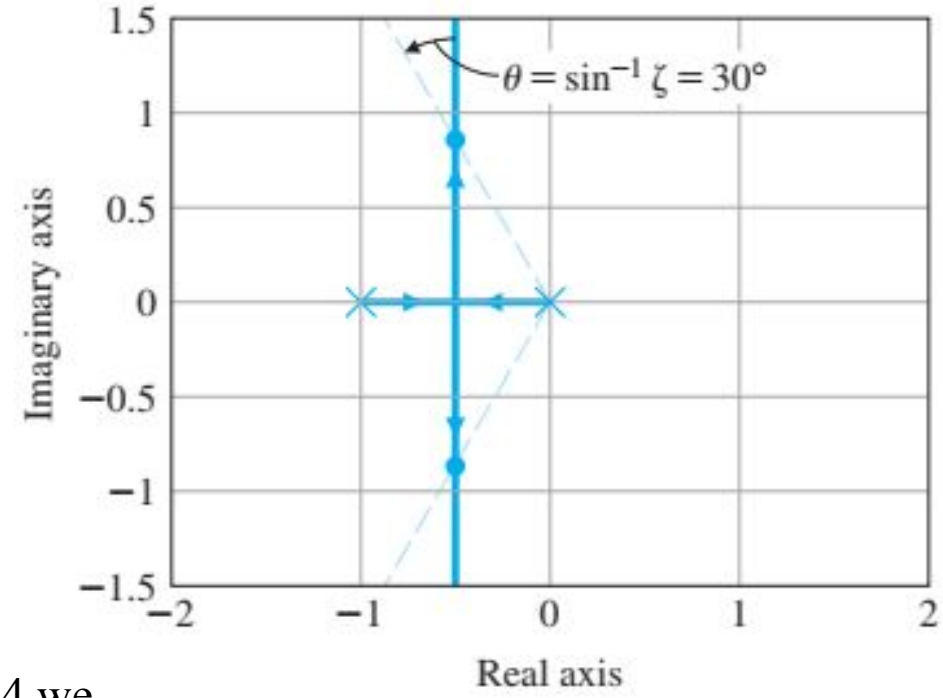
Guidelines for Determining a Root Locus:

Rule 5: Example

- $K_1 = 1/4$
- $K = 1/4 + K_2$

$$s^2 + s + 1/4 + K_2 = 0$$

$$\left(s + \frac{1}{2}\right)^2 + K_2 = 0$$



$$a(s) + Kb(s) = s^2 + s + K = 0.$$

$$r_1, r_2 = -\frac{1}{2} \pm \frac{\sqrt{1 - 4K}}{2}$$

Using the equation of Rule 4 we calculate the "departure" angles

$$q\phi_{l,dep} = \sum \psi_i - \sum_{i \neq l,dep} \phi_i - 180° - 360°(l - 1).$$

2. Guidelines for Determining a Root Locus

Guidelines for Determining a Root Locus:

Rule 5: Example

- $K_1 = 1/4$
- $K = 1/4 + K_2$

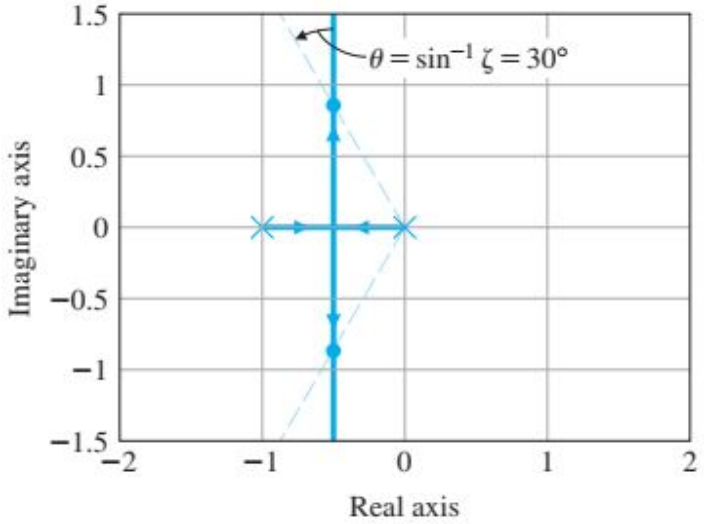
$$2\phi_{dep} = -180° - 360°(l - 1),$$
$$\phi_{dep} = -90° - 180°(l - 1),$$
$$\phi_{dep} = \pm 90° \text{ (departure angles at breakaway)}.$$

$$s^2 + s + 1/4 + K_2 = 0$$

$$\left(s + \frac{1}{2}\right)^2 + K_2 = 0$$

$$a(s) + Kb(s) = s^2 + s + K = 0.$$

$$r_1, r_2 = -\frac{1}{2} \pm \frac{\sqrt{1 - 4K}}{2}$$



Using the equation of Rule 4 we calculate the "departure" angles

$$q\phi_{l,dep} = \sum \psi_i - \sum_{i \neq l,dep} \phi_i - 180° - 360°(l - 1).$$

2. Guidelines for Determining a Root Locus

Summary

- RULE 1. The n branches of the locus start at the poles of L(s) and m branches end on the zeros of L(s).
- RULE 2. The loci are on the real axis to the left of an odd number of poles and zeros.
- RULE 3. For large s and K, n − m branches of the loci are asymptotic to lines at angles φ_l radiating out from the center point s = α on the real axis (see (1) and (2).
- RULE 4. The angle(s) of departure of a branch of the locus from a pole of multiplicity q is given by (3), where l = 1, 2, ... , q and the angle(s) of arrival of a branch at a zero of multiplicity q is given by (4).
- RULE 5. The locus can have multiple roots at points on the locus of multiplicity q. The branches will approach to/depart from a point of q roots at angles separated by the angles given in (5), where l = 1, 2, ... , q

$$\phi_l = \frac{180^\circ + 360^\circ(l-1)}{n-m} \quad (1)$$

$$\alpha = \frac{\sum p_i - \sum z_i}{n-m}. \quad (2)$$

$$l = 1, 2, \ldots, n-m,$$

$$q\phi_{l,dep} = \sum \psi_i - \sum \phi_i - 180^\circ - 360^\circ(l-1) \quad (3)$$

$$l = 1, 2, \ldots, q$$

$$q\psi_{l,arr} = \sum \phi_i - \sum \psi_i + 180^\circ + 360^\circ(l-1) \quad (4)$$

$$l = 1, 2, \ldots, q$$

$$\frac{180^\circ + 360^\circ(l-1)}{q} \quad (5)$$

$$l = 1, 2, \ldots, q$$

2. Guidelines for Determining a Root Locus

Selecting the Parameter Value

- **Option 1:** using Matlab (example)

```
s=tf('s');
G=1/(s*((s+4)^2 + 16));
rlocus(G);


[K,p] = rlocfind(G);


sisotool('rlocus',G);
```

2. Guidelines for Determining a Root Locus

Selecting the Parameter Value

- **Option 2:** from L(s)

$$L(s) = \frac{1}{s[(s+4)^2 + 16]}$$

Characteristic equation

$$K = -\frac{1}{L(s)} \qquad K = \frac{1}{|L|}$$

Drawing lines from the **poles and zeros of L(s)** to the evaluation point $\mathbf{s_0}$

|L(s)| can be calculated by calculation or by the root locus plot with simetrical axes

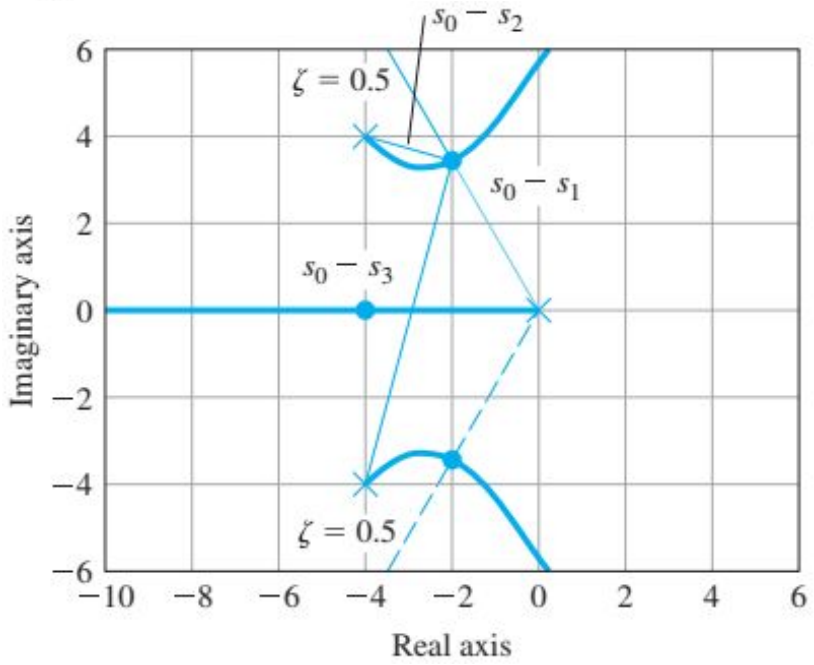
$$L(s_0) = \frac{1}{s_0(s_0 - s_2)(s_0 - s_3)} \qquad K = \frac{1}{|L(s_0)|} = |s_0||s_0 - s_2||s_0 - s_3|$$

$$|s_0| \cong 4.0,$$

$$|s_0 - s_2| \cong 2.1,$$

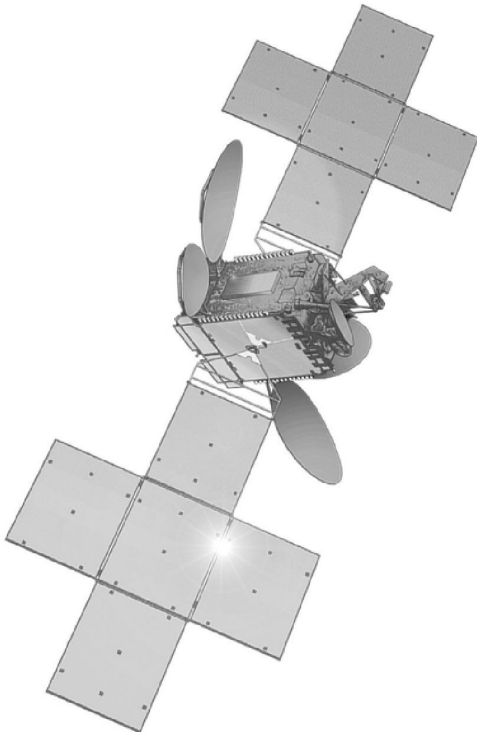
$$|s_0 - s_3| \cong 7.7.$$

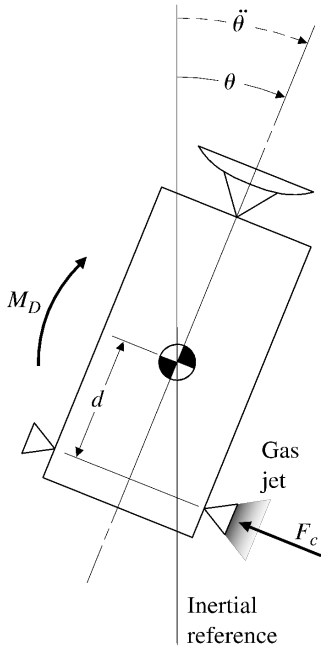
$$K = 4.0(2.1)(7.7) \cong 65$$



3. Selected Illustrative Root Loci

Example: Root Locus for Satellite Attitude Control with PD Control





$$F_c d + M_D = I\ddot{\theta}$$

To obtain θ => integrate the sum of the input torques twice (**double-integrator plant**)

Substituting $U = F_{cd} + M_D$ and s for d/dt

$$\frac{\Theta(s)}{U(s)} = \frac{1}{I}\frac{1}{s^2}$$

Assuming I = 1

3. Selected Illustrative Root Loci

Example: Root Locus for Satellite Attitude Control with PD Control

- We define $K = k_D$
- We arbitrarily select the gain ratio as $k_p/k_D = 1$ (this ratio can be defined depending on the rise time)

$$G(s) = \frac{1}{s^2}$$

Characteristic equation (closed-loop)

$$1 + [k_p + k_D s]\frac{1}{s^2} = 0.$$

Characteristic equation (closed-loop) with $k_p/k_D = 1$

$$1 + K\frac{s+1}{s^2} = 0$$

3. Selected Illustrative Root Loci

Example: Root Locus for Satellite Attitude Control with PD Control

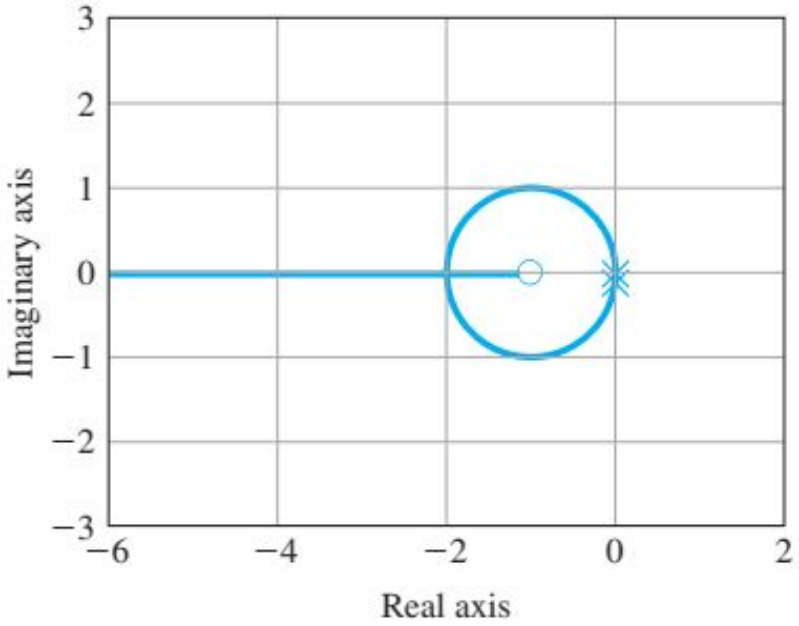
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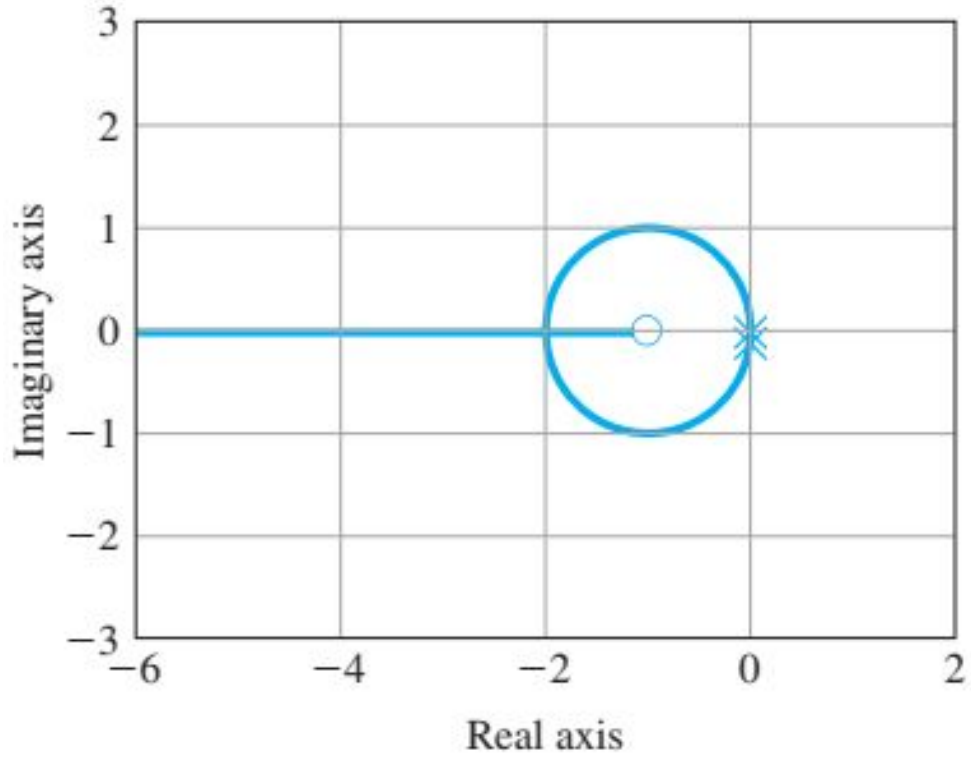
$$1 + K\frac{s+1}{s^2} = 0$$



3. Selected Illustrative Root Loci

Example: Root Locus for Satellite Attitude Control with PD Control

- The addition of the zero has **pulled** the **locus** into the **LHP**
- **I**mportant in constructing a compensation



3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

- To implement the PD it is necessary to add a pole **p**
- **p** need to be far from the poles and zeros of the plant and the controller

$$D_c(s) = k_p + \frac{k_D s}{s/p + 1}$$

- Defining

$$K = k_p + pk_D \qquad z = pk_p/K$$

$$D_c(s) = K\frac{s+z}{s+p}$$

$$1 + D_c(s)G(s) = 1 + KL(s) = 0$$

$$1 + K\frac{s+z}{s^2(s+p)} = 0.$$

Lets analyze the following cases:
p = 12 (pole at s = -12)
p = 4 (pole at s = -4)
p = 9 (pole at s = -9)

3. Selected Illustrative Root Loci

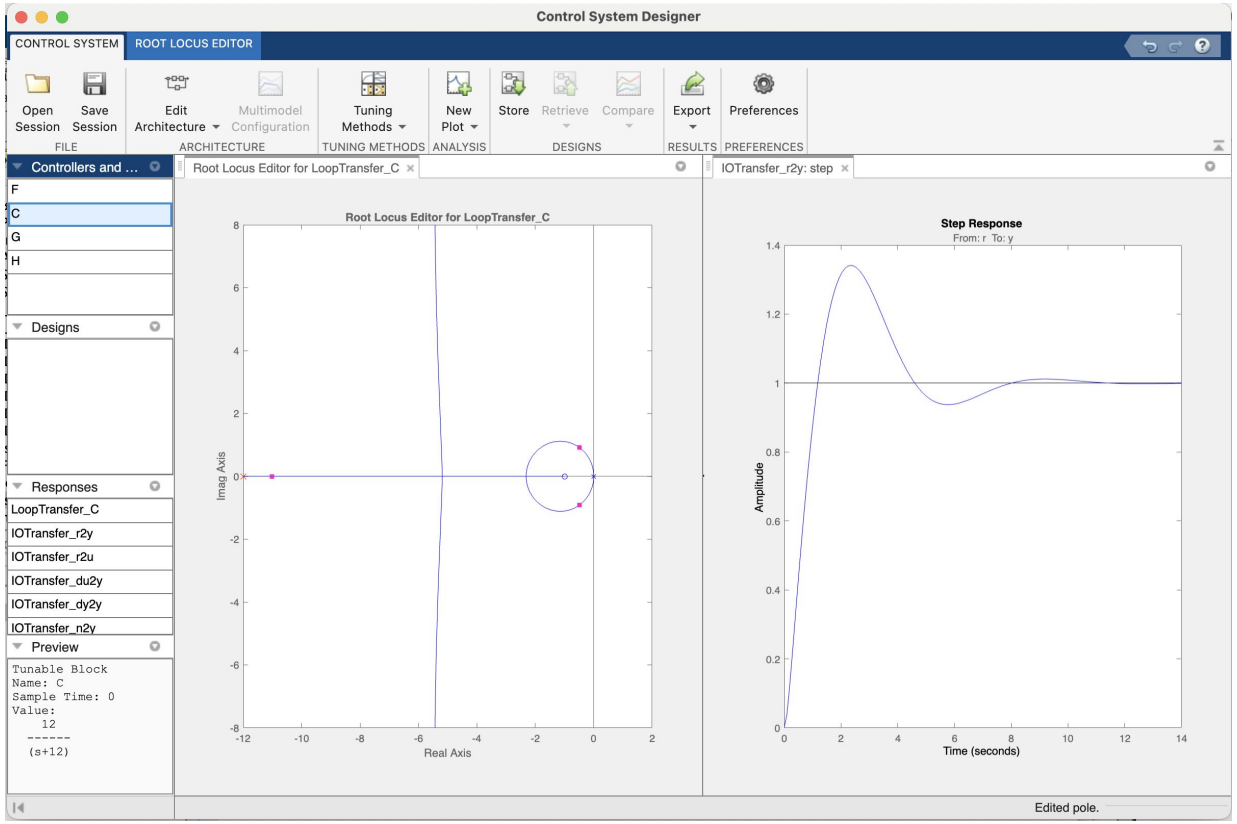
Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

With z = 1

p = 12 (pole at s = -12)
p = 4 (pole at s = -4)
p = 9 (pole at s = -9)

```
s=tf('s');
G=(s+1)/(s^2);

sisotool('rlocus',G);
```



3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

p = 12 (pole at s = -12)

$$1 + K\frac{s+1}{s^2(s+12)} = 0$$

RULE 1. There are now three branches to the locus, two starting at $s = 0$ and one starting at $s = -12$.

RULE 2. The real axis segment $-12 \leq s \leq -1$ is part of the locus.

RULE 3. There are $n - m = 3 - 1 = 2$ asymptotes centered at $\alpha = \frac{-12-(-1)}{2} = -11/2$ and at the angles $\pm 90°$.

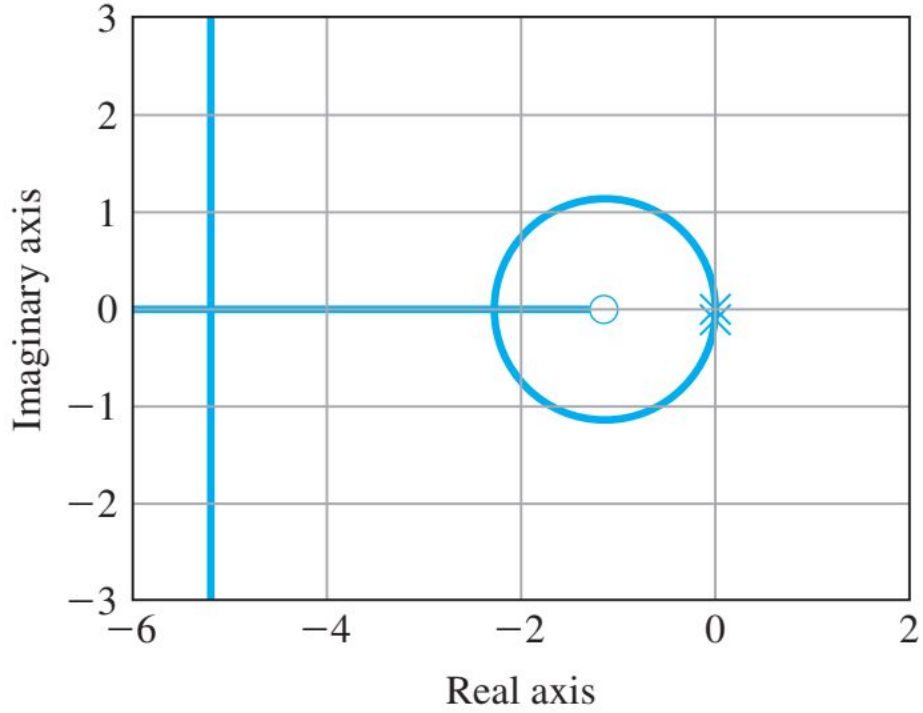
RULE 4. The angles of departure of the branches at $s = 0$ are again $\pm 90°$. The angle of departure from the pole at $s = -12$ is at $0°$.

3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

p = 12 (pole at s = -12)

$$1 + K\frac{s+1}{s^2(s+12)} = 0$$



3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

p = 4 (pole at s = -4)

$$1 + K\frac{s+1}{s^2(s+4)} = 0$$

RULE 1. There are again three branches to the locus, two starting from $s = 0$ and one from $s = -4$.

RULE 2. The segment of the real axis $-4 \leq s \leq -1$ is part of the locus.

RULE 3. There are two asymptotes centered at $\alpha = -3/2$ and at the angles $\pm 90°$.

RULE 4. The branches again depart from the poles at $s = 0$ at $\pm 90°$.

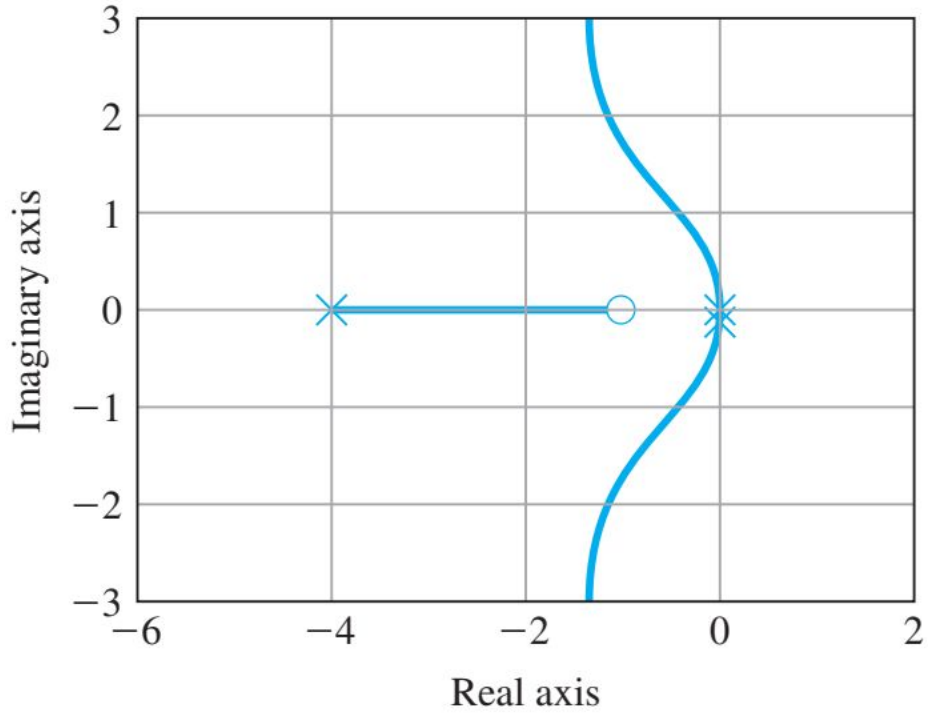
RULE 5. The Matlab commands

3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

p = 4 (pole at s = -4)

$$1 + K\frac{s+1}{s^2(s+4)} = 0$$



3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

p = 12 (pole at s = -12)

$$1 + K\frac{s+1}{s^2(s+9)} = 0$$

RULE 1. The locus has three branches, starting from $s = 0$ and $s = -9$

RULE 2. The real axis segment $-9 \leq s \leq -1$ is part of the locus.

RULE 3. The two asymptotes are centered at $\alpha = -8/2 = -4$.

RULE 4. The departures are, as before, at $\pm 90°$ from $s = 0$.

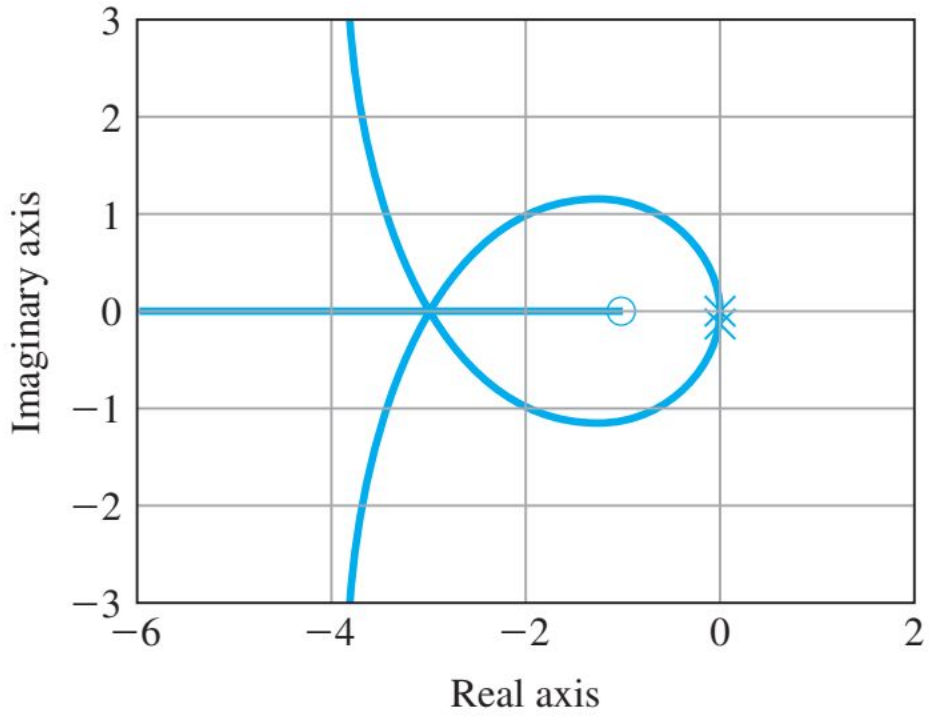
RULE 5. The Matlab commands

3. Selected Illustrative Root Loci

Example: Root Locus of the Satellite Control with Modified PD or Lead Compensation

p = 9 (pole at s = -9)

$$1 + K\frac{s+1}{s^2(s+9)} = 0$$



4. Design Using Dynamic Compensation

1. Determine where the closed-loop roots need to be in the s-plane in order to meet the desired specifications on the speed of response and damping (or overshoot).
   (a) pick the limits for ω_n and ζ, or
   (b) pick the limits for σ and ω_d.
2. Create the root locus vs. K with no compensation.
3. If more damping is required, select a value of z in Eq. (5.70) to be approximately 1/4 to 1 times the value of the desired ω_n or ω_d and pick p to be $10*z$.
4. Examine the resulting root locus; and adjust as necessary to meet the required specifications as determined in step 1.
   (a) decrease p if less damping is needed,
   (b) increase p if more damping is needed, and/or decrease z,
   (c) it is desirable to keep the value of p/z as low as possible ($p/z \lesssim 25$) in order to minimize the amplification of sensor noise by the compensation.
5. When the values of z and p are selected so that the resulting locus passes through an acceptable region of the s-plane, determine the value of K to select the closed-loop root locations.
6. Verify that all time domain specifications are met by examining the time response to a unit step input, and adjust the desired s-plane specifications if needed and go back to step 2.
7. Determine if the resulting value of K meets the steady-state error requirements, if any. If a value of K can not be found that meets the requirement, then add Integral Control or a Lag Compensator.

Design of lead compensator

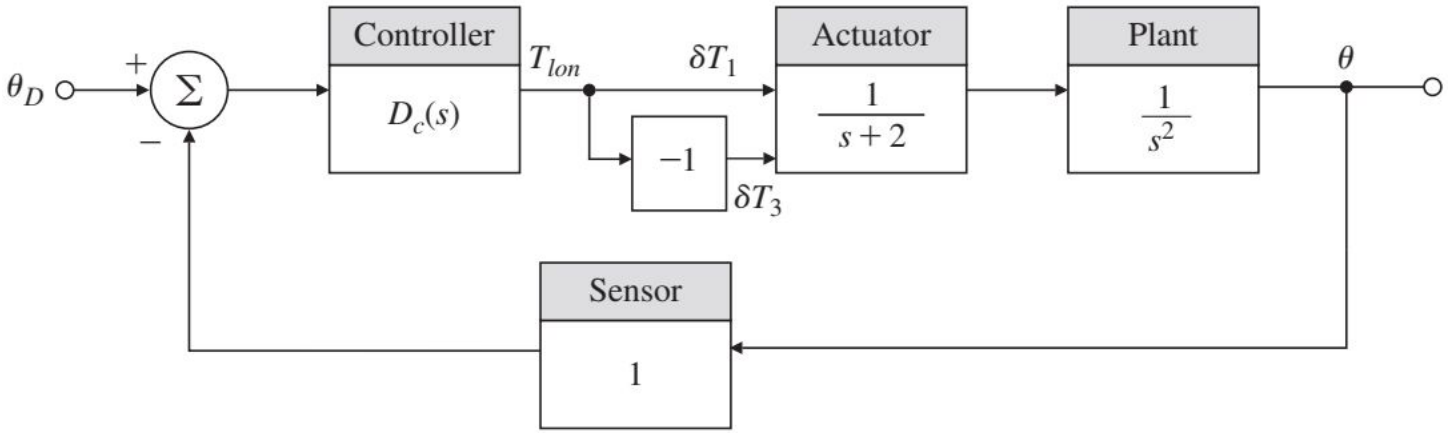
4. Design Using Dynamic Compensation

1. Determine the amount of gain amplification to be contributed by the lag compensation at low frequencies in order to achieve the desired K_p or K_v or K_a as determined by Eqs. (4.36–4.38).
2. Select the value of z in Eq. (5.72) so it is approximately a factor of 100 to 200 smaller than the system dominant natural frequency.
3. Select the value of p in Eq. (5.72) so that z/p is equal to the desired gain amplification determined in step 1.
4. Examine the resulting root locus to verify that the frequency and damping of the dominant closed-loop roots are still satisfactory. If not, adjust the lead compensation as needed.
5. Verify that all time domain specifications are met by examining the time response to a unit step input. If the slow root introduced by the lag compensation is too slow, increase the values of z and p somewhat while keeping z/p constant, and go back to step 4. However, do so with the understanding that the closer the values of the lag compensator's z and p come to the dominant roots of the closed loop system, the more they will affect those dominant root characteristics.

Design of lag compensator

5. Design Examples Using the Root Locus

Control of a Quadrotor Drone Pitch Axis



$$\frac{\theta(s)}{T_{lon}(s)} = G_1(s) = \frac{1}{s^2(s+2)}$$

The lag term (s + 2) is for quadrotor of 2 meters in diameter, aprox.

Desired specifications:

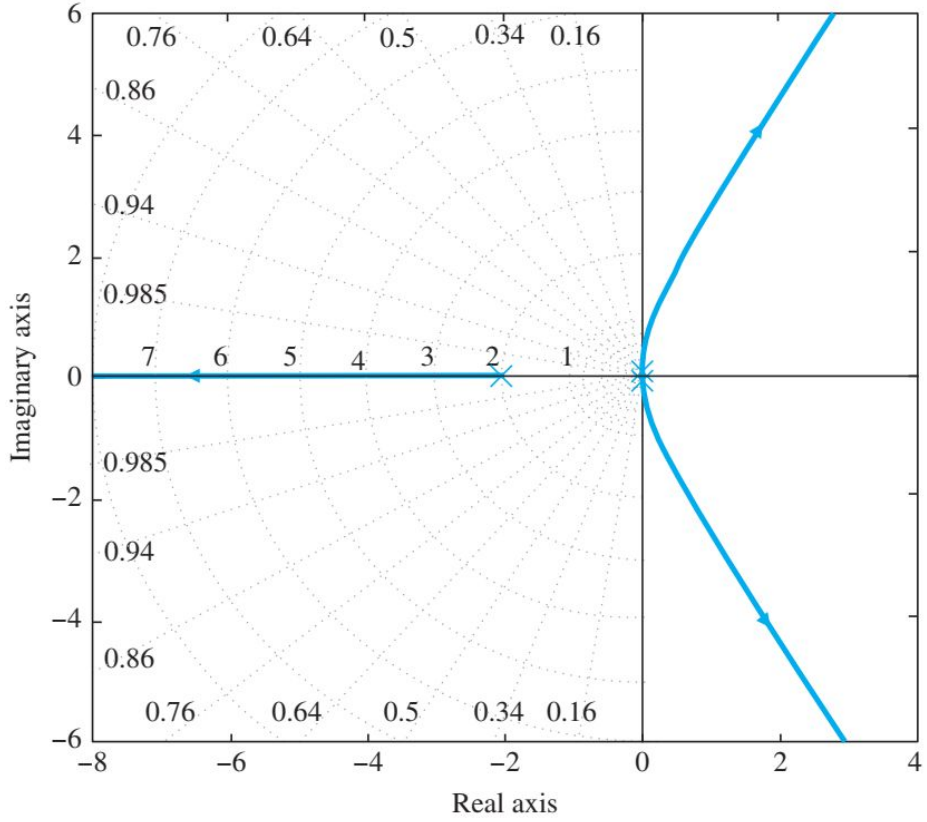
$$\omega_n \geq 1 \text{ rad/sec}$$

$$\zeta \geq 0.44.$$

Compensator: lead

5. Design Examples Using the Root Locus

Control of a Quadrotor Drone Pitch Axis



$\omega_n \geq 1$ rad/sec

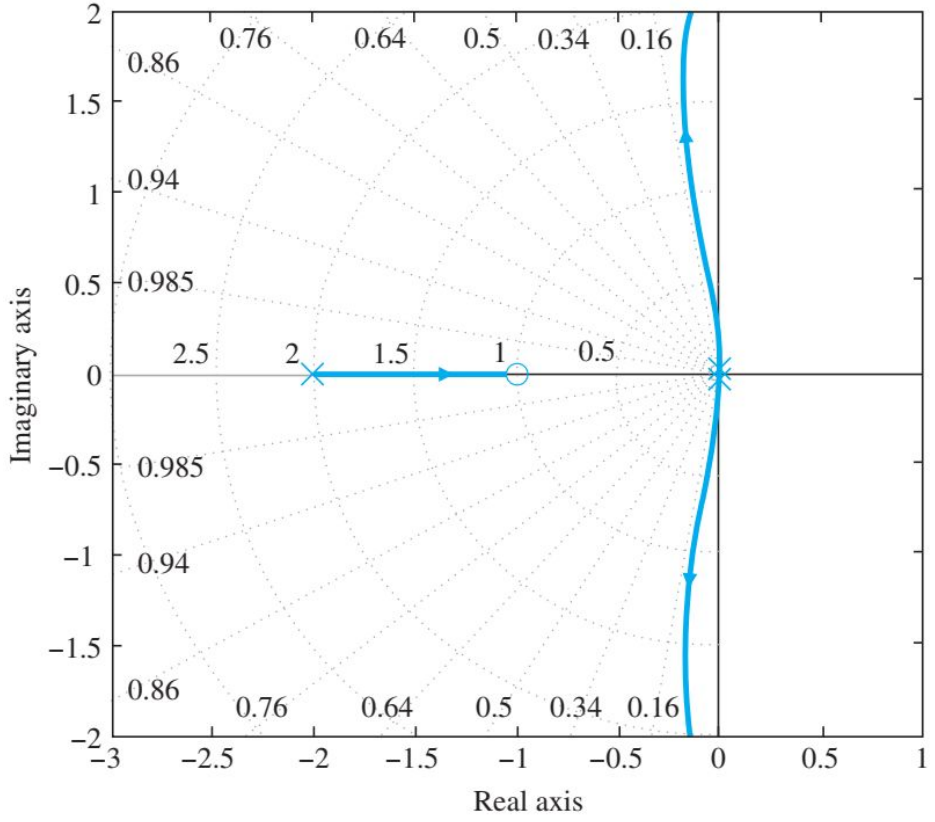
Step 1: specifications $\zeta \geq 0.44$.

Step 2: Root locus in open-loop

```
s = tf('s');
sysG1=1/((s^2)*(s + 2));
rlocus(sysG1);
axis([-8 4 -6 6]);
grid on;
```

5. Design Examples Using the Root Locus

Control of a Quadrotor Drone Pitch Axis



$\omega_n \geq 1$ rad/sec

Step 1: specifications $\zeta \geq 0.44.$

Step 2: Root locus in open-loop

Step 3: lead compensator with z = 1 and p = 10 and generate the root locus

$$D_c(s) = K\frac{s+1}{s+10}$$

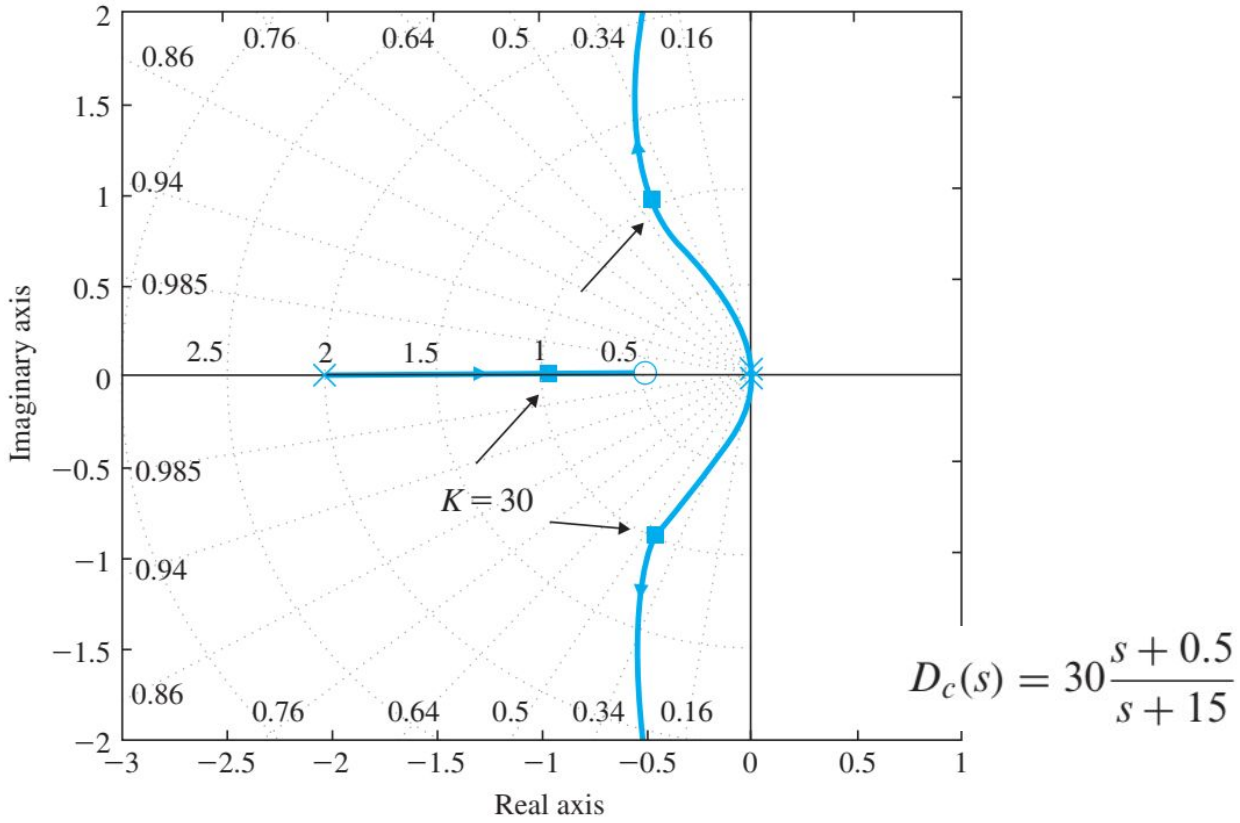
```
sysD=(s+1)/(s+10);
figure(2);
rlocus(sysG1*sysD);
axis([-3 1 -2 2]);
grid on;
```

5. Design Examples Using the Root Locus

Control of a Quadrotor Drone Pitch Axis

$\omega_n \geq 1$ rad/sec

Step 1: specifications $\zeta \geq 0.44.$



$$D_c(s) = 30\frac{s+0.5}{s+15}$$

Step 2: Root locus in open-loop

Step 3: lead compensator with z = 1 and p = 10 and generate the root locus => no feasible

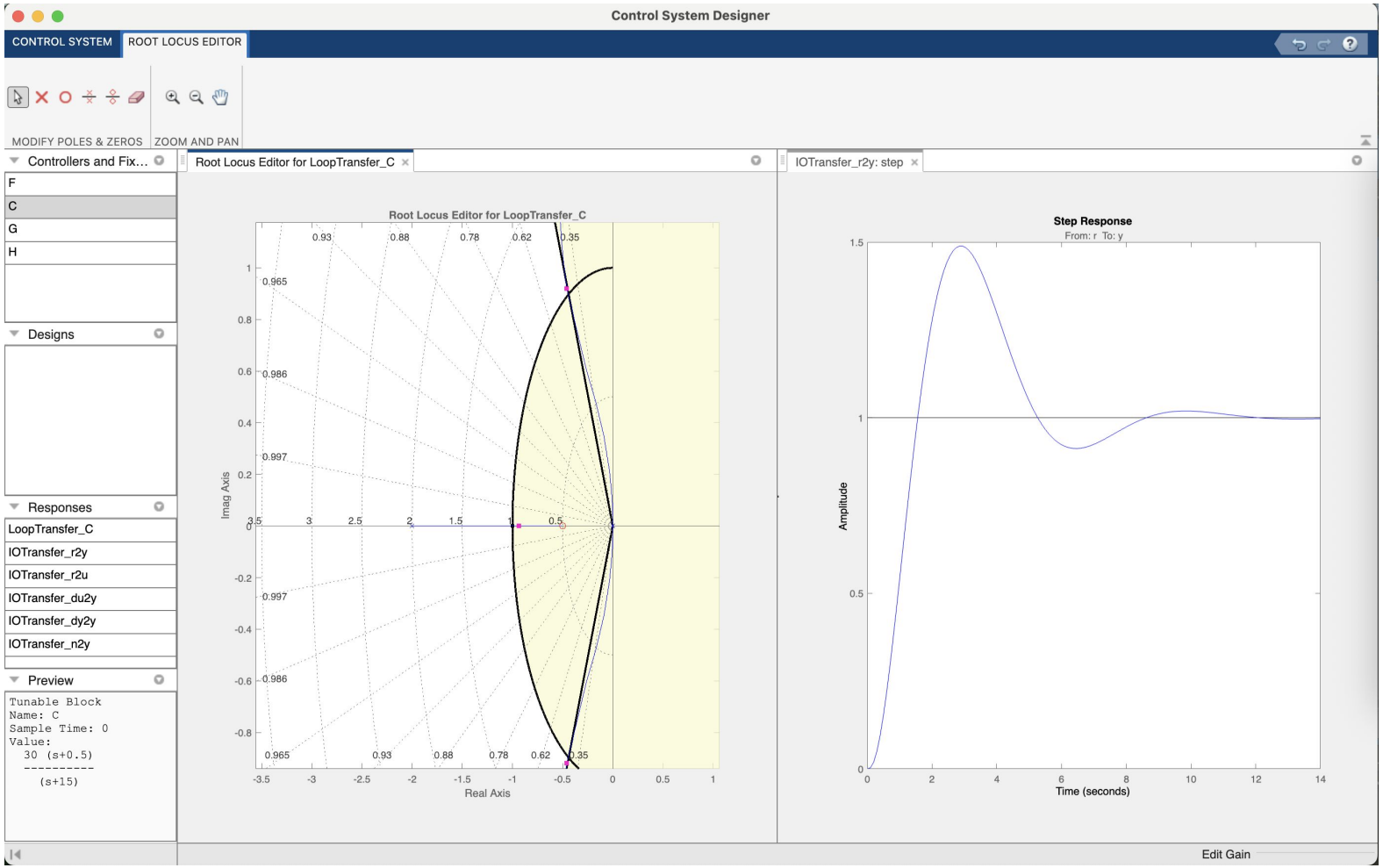
Step 4: lead compensator with z = 0.5 and p = 15

Step 5: verify (using the cursor in Matlab's plot)

5. Design Examples Using the Root Locus

Control of a Quadrotor Drone Pitch Axis

```
Kctrl = 30;
sysD = Kctrl*sysD;
sisotool('rlocus',sysG1,sysD);
```



5. Design Examples Using the Root Locus

Control of a Small Airplane

Transfer function between the elevator input and the pitch attitude

$$G(s) = \frac{\theta(s)}{\delta_e(s)} = \frac{160(s + 2.5)(s + 0.7)}{(s^2 + 5s + 40)(s^2 + 0.03s + 0.06)}$$

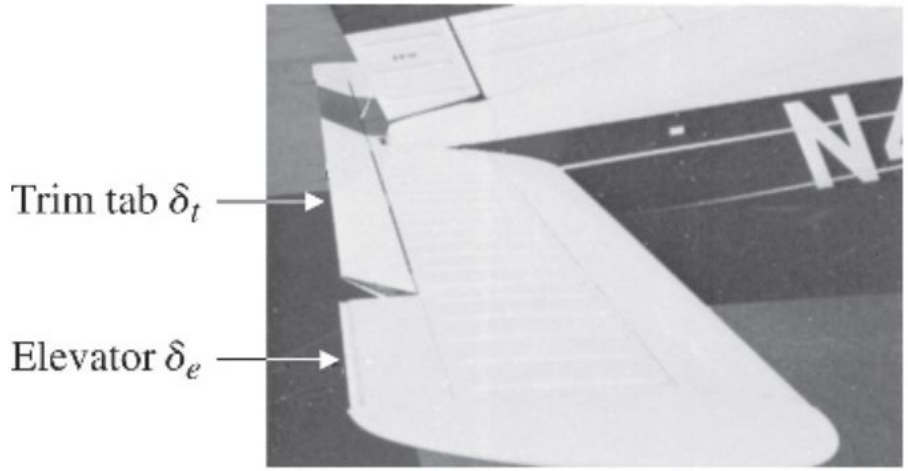
θ = pitch attitude, degrees

δ_e = elevator angle, degrees.

$$\frac{\theta(s)}{M_d(s)} = \frac{160(s + 2.5)(s + 0.7)}{(s^2 + 5s + 40)(s^2 + 0.03s + 0.06)}$$

M_d is the moment acting on the aircraft





5. Design Examples Using the Root Locus

Control of a Small Airplane

Transfer function between the elevator input and the pitch attitude

$$G(s) = \frac{\theta(s)}{\delta_e(s)} = \frac{160(s + 2.5)(s + 0.7)}{(s^2 + 5s + 40)(s^2 + 0.03s + 0.06)}$$

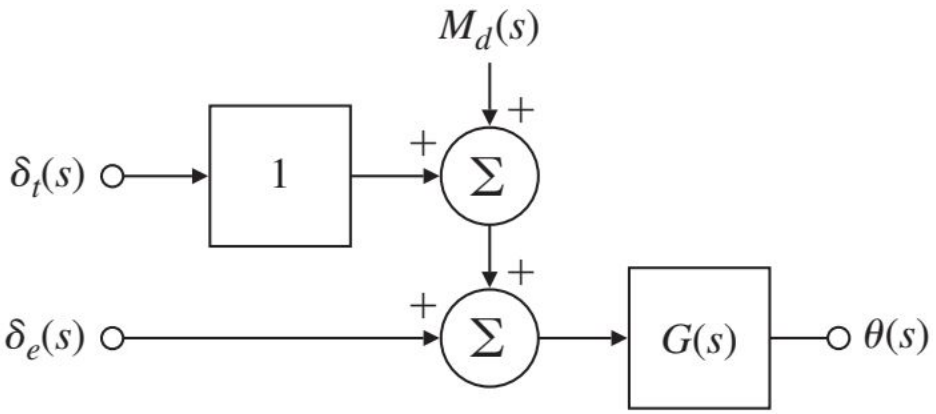
θ = pitch attitude, degrees

δ_e = elevator angle, degrees.

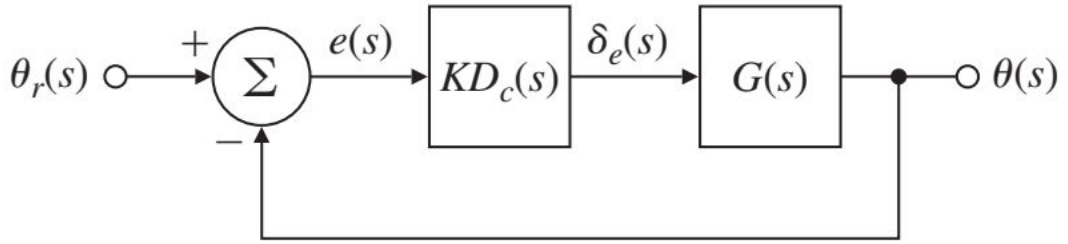
$$\frac{\theta(s)}{M_d(s)} = \frac{160(s + 2.5)(s + 0.7)}{(s^2 + 5s + 40)(s^2 + 0.03s + 0.06)}$$

M_d is the moment acting on the aircraft

Open-loop



Feedback scheme excluding trim



5. Design Examples Using the Root Locus

Control of a Small Airplane

Transfer function between the elevator input and the pitch attitude

$$G(s) = \frac{\theta(s)}{\delta_e(s)} = \frac{160(s + 2.5)(s + 0.7)}{(s^2 + 5s + 40)(s^2 + 0.03s + 0.06)}$$

θ = pitch attitude, degrees

δ_e = elevator angle, degrees.

$$\frac{\theta(s)}{M_d(s)} = \frac{160(s + 2.5)(s + 0.7)}{(s^2 + 5s + 40)(s^2 + 0.03s + 0.06)}$$

M_d is the moment acting on the aircraft

1. Design an autopilot so the response to a step elevator input has a rise time of 1 sec or less and an overshoot less than 10%.

2. Design an autopilot that will command the trim δt so as to drive the steadystate value of δe to zero for an arbitrary constant moment Md as well as meet the specifications in part (a).

5. Design Examples Using the Root Locus

Control of a Small Airplane

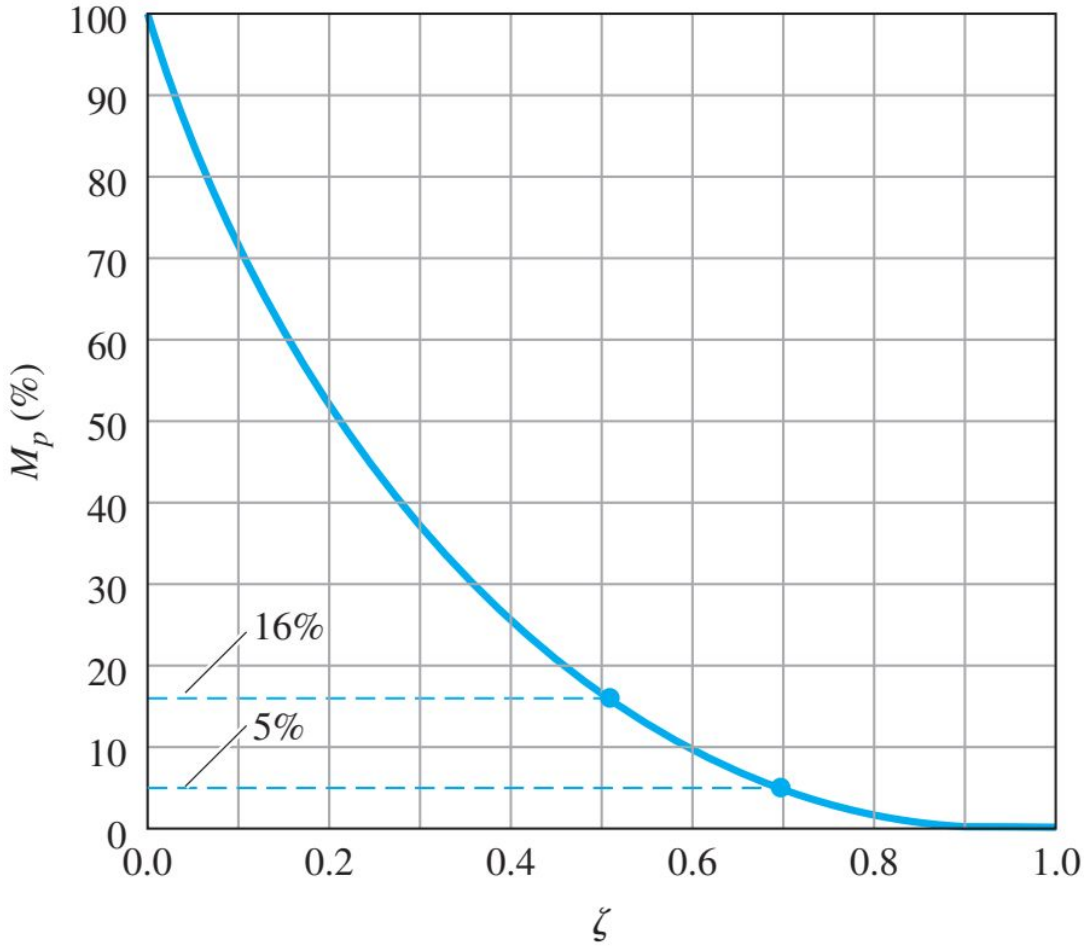
Step 1: design requirements

1. Design an autopilot so the response to a step elevator input has a rise time of 1 sec or less

$$t_r \cong \frac{1.8}{\omega_n}, \quad \omega_n > 1.8 \text{ rad/s}$$

and an overshoot less than 10%.

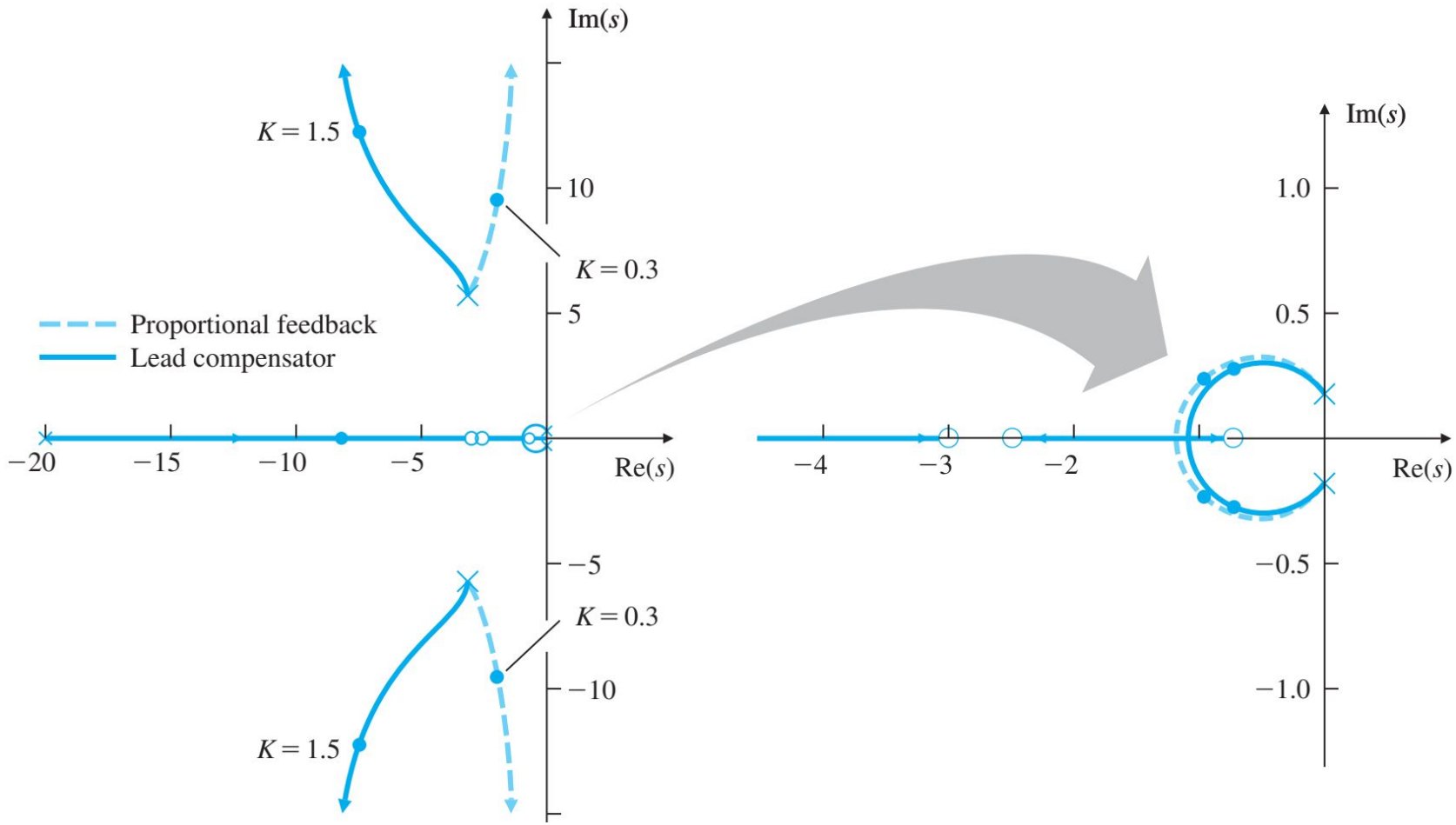
From the figure: $\zeta > 0.6$



5. Design Examples Using the Root Locus

Control of a Small Airplane

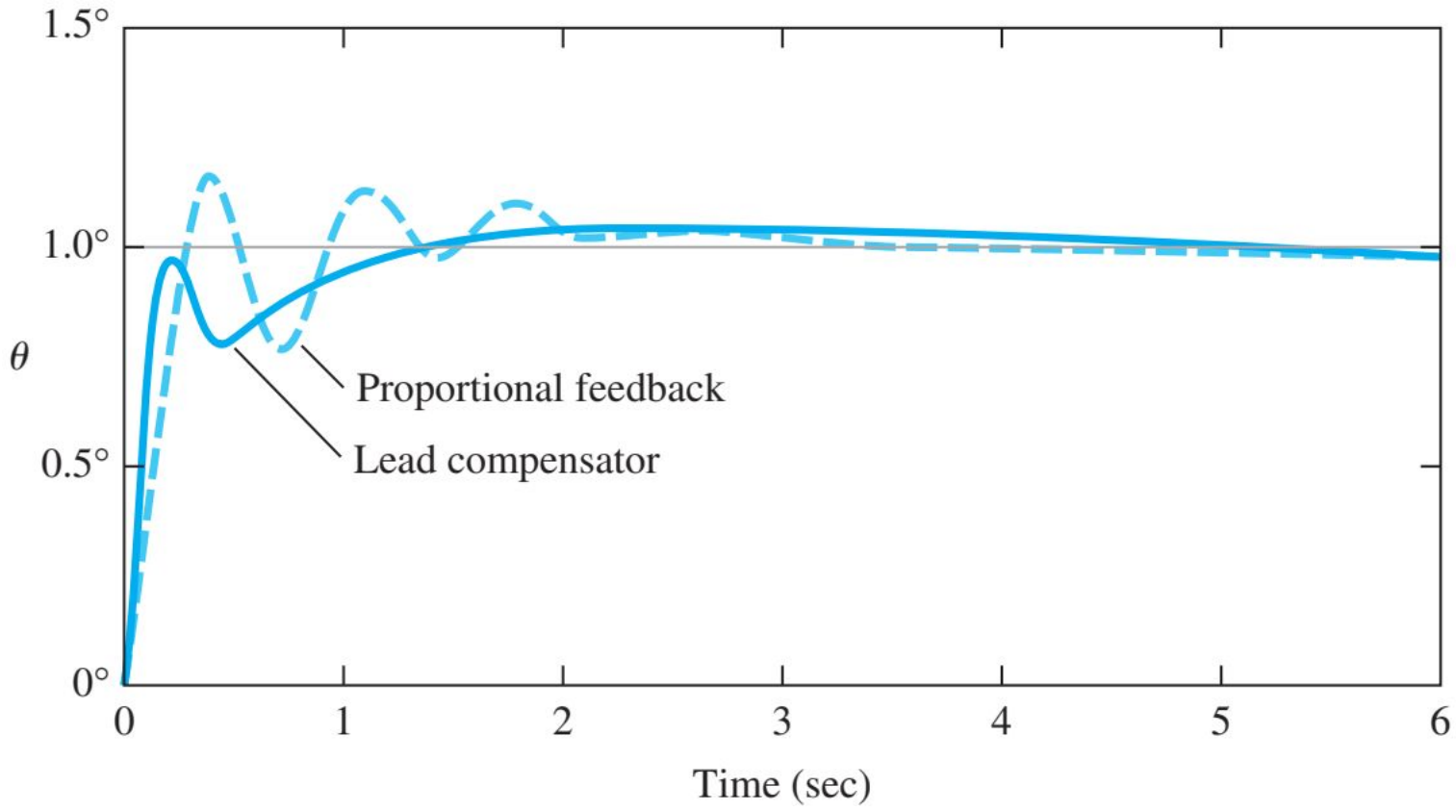
Step 2: root locus in open-loop (dashed lines)



5. Design Examples Using the Root Locus

Control of a Small Airplane

Step 2: step response in open-loop (dashed lines)



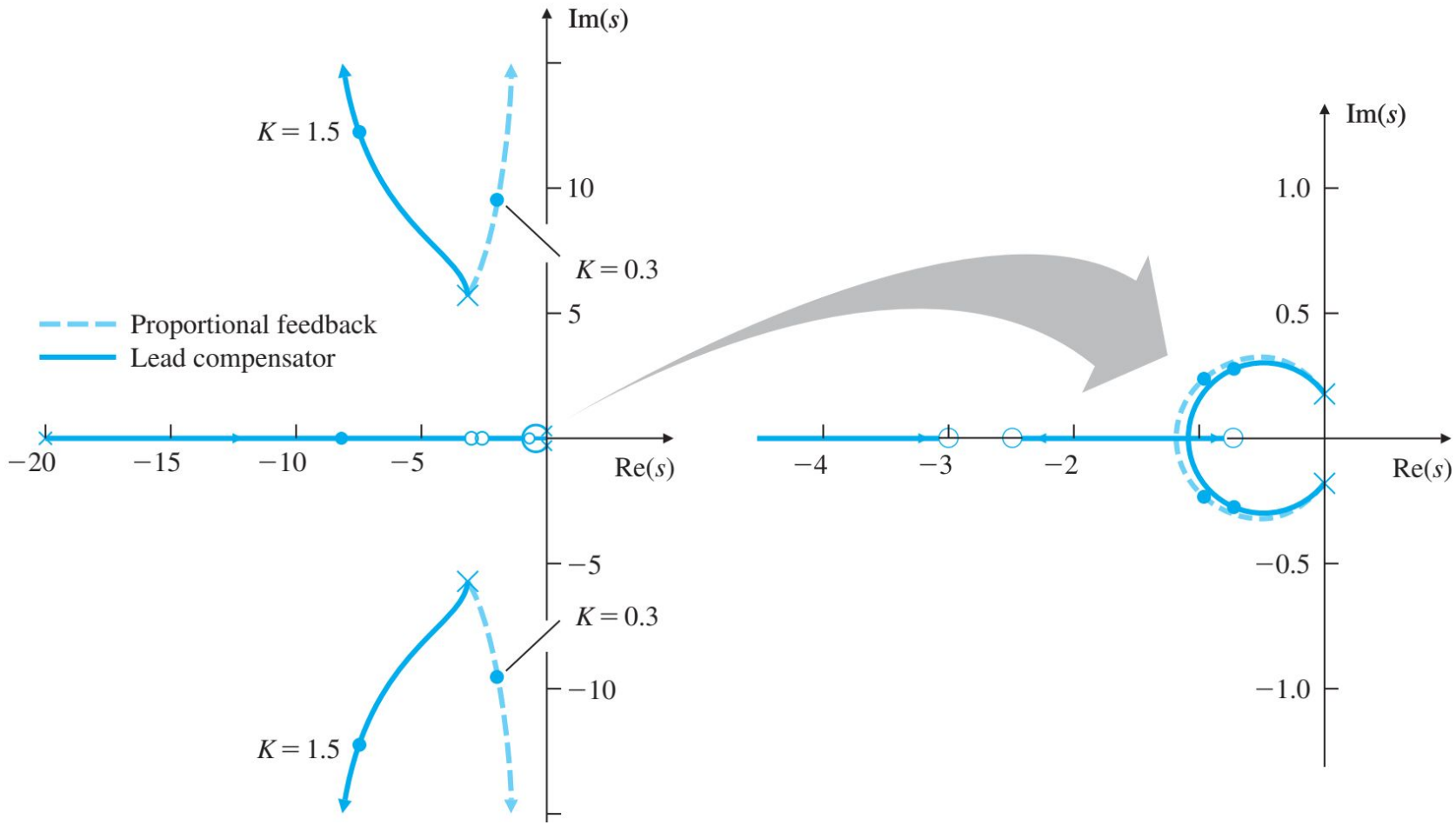
5. Design Examples Using the Root Locus

Control of a Small Airplane

Step 3: try lead compensator with z = 3 and p = 20

$$D_c(s) = \frac{s+3}{s+20}$$

(solid lines)



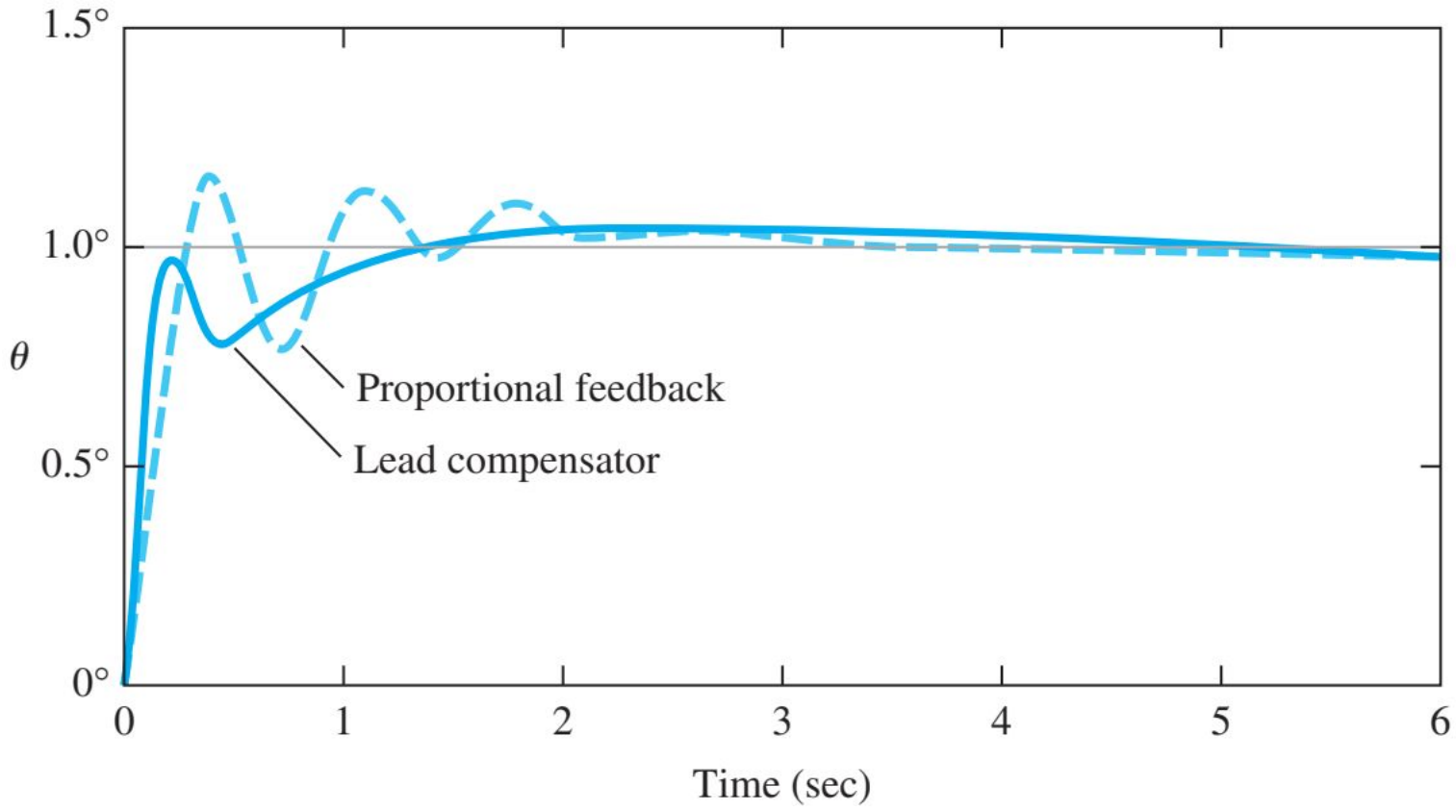
5. Design Examples Using the Root Locus

Control of a Small Airplane

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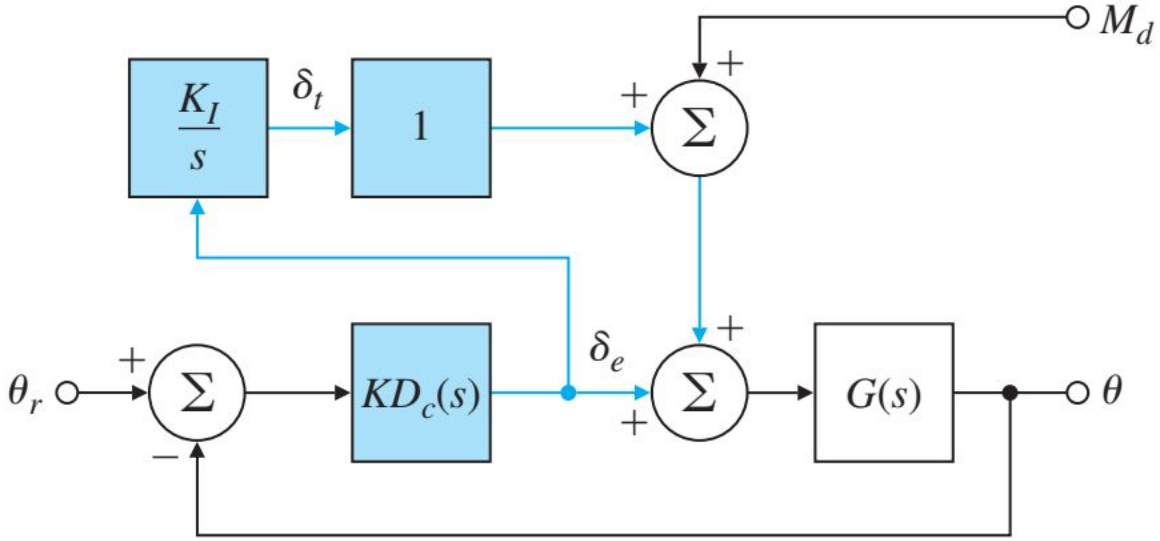
5. Design Examples Using the Root Locus

Control of a Small Airplane

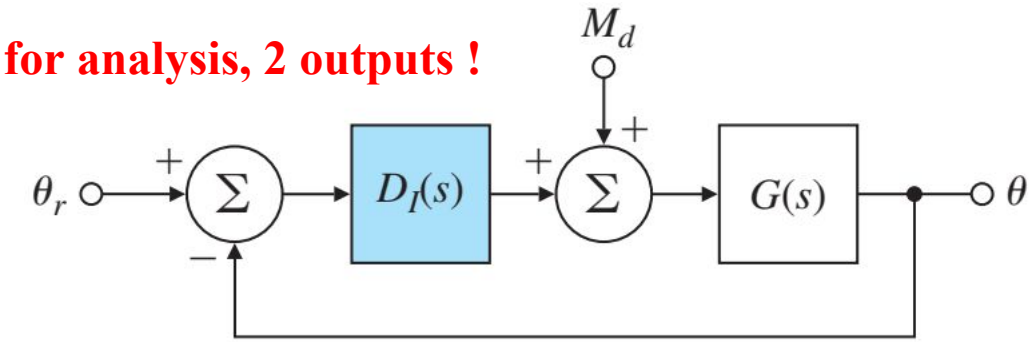
2. Design an autopilot that will command the trim δt so as to drive the steadystate value of δe to zero for an arbitrary constant moment Md as well as meet the specifications in part (a).

Idea: take δe to 0 with an I controller

$$D_I(s) = KD_c(s)\left(1 + \frac{K_I}{s}\right)$$



Just for analysis, 2 outputs !



5. Design Examples Using the Root Locus

Control of a Small Airplane

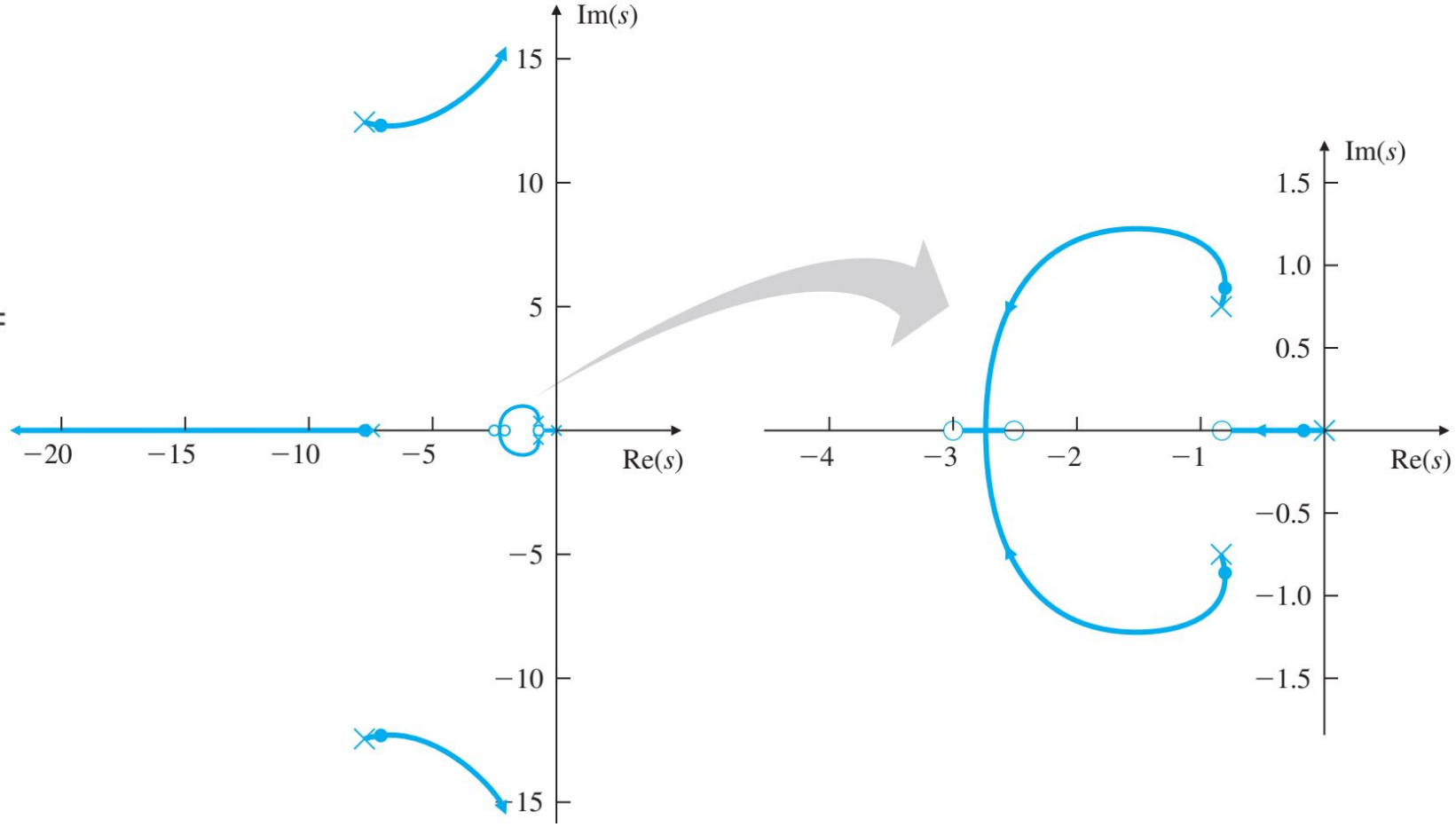
Root locus with respect to K_I

$$1 + KD_cG + \frac{K_I}{s}KD_cG =$$

Dividing by (1 + KDcG)

$$1 + \frac{(K_I/s)KD_cG}{1 + KD_cG} = 0$$

$$L(s) = \frac{1}{s}\frac{KD_cG}{1 + KD_cG}$$



5. Design Examples Using the Root Locus

Control of a Small Airplane

Keep K_I as low as possible

$K_I = 1$ to obtain approximately the same settling time

The steady-state is 0

Remember: 2 outputs !